Jomo Kenyatta University of Agriculture and Technology

EARLY DIAGNOSIS OF PNEUMONIA FROM CHEST X-RAYS USING A CAPSULE NETWORK MODEL: ENHANCING ACCURACY AND EFFICIENCY IN AUTOMATED IMAGE CLASSIFICATION

A Research Project Submitted in Partial Fulfillment of the Requirements for Award of th

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DECLARATION

This project is my original work and has not been presented elsewhere for the award of a degree in any other institution. I, therefore, declare all material cited in this write-up that is not mine has been duly acknowledged.

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ABSTRACT

Pneumonia is a significant public health concern worldwide, causing substantial morbidity and mortality. Early diagnosis is vital in ensuring timely treatment and improving patient outcomes. The most common diagnosis method for pneumonia is analysing chest X-rays, but the accurate and timely interpretation of these images can be challenging. This research aimed to develop a capsule network (CapsNet) model for image classification, enabling automated analysis of chest X-rays for early diagnosis of pneumonia.

Pneumonia affects millions worldwide, particularly the elderly, young children, and individuals with weakened immune systems. Delayed or missed diagnoses can lead to severe complications and increased healthcare costs. However, accurately interpreting chest X-rays, commonly used for pneumonia diagnosis, requires specialized expertise and can be prone to human error. Therefore, developing automated and accurate diagnostic models is crucial for timely interventions. The model was trained on secondary data obtained from Mendeley data. The data was pre-processed thoroughly using multiple techniques to capture the main features of the pneumonia-infected images to develop a model favorable for early diagnosis. The model achieved a test accuracy of 0.9678 with a loss of 0.3265, a precision of 0.8214, and a recall of 0.8423. Hashmi et al. (2020)

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Chapter 1

INTRODUCTION

1.1 Background of the study

Pneumonia is defined by the World Health Organization (WHO) as an acute respiratory illness that affects the lungs, whereby the alveoli in the lungs are stuffed with pus and fluid instead of air (WHO,2022). Pneumonia patients experience breathing difficulties, and their ability to take in oxygen is restricted drastically. Pneumonia is caused by several infectious organisms, including bacteria, fungi, and viruses; the most frequent ones are Streptococcus pneumoniae, Haemophilus influenzae type b, Respiratory syncytial virus, which is the most prevalent viral cause of pneumonia, and Pneumocystis jiroveci (WHO, 2022). The Centers for Disease Control and Prevention(CDC) classify pneumonia as a lower respiratory tract infection with bronchitis and Tuberculosis. Lower respiratory tract diseases were listed as the fourth leading cause of mortality worldwide in the WHO 2020 fact sheet. Over 450 million individuals suffer from pneumonia yearly Ruuskanen et al. (2011). Ruuskanen et al. (2011), further described that pneumonia causes more than four million deaths in a year, accounting for 7% of the overall population around the globe. Pneumonia is the single largest infectious cause of death in children worldwide; pneumonia killed 740 180 children under the age of 5 in 2019, accounting for 14% of all deaths of children under five years old and 22% of all deaths in children aged 1 to 5,(WHO,2021). These figures indicate that pneumonia affects children and families everywhere and is a global pandemic that needs intervention. According to the World Health Organization (WHO), pneumonia is responsible for millions of deaths annually, making it a significant global health concern.

Mokdad et al. (2018) in their study *"Burden of diseases, injuries, and risk factors among US states"* state that pneumonia morbidity and mortality remain high and inequitably distributed, with deaths from pneumonia concentrated in Sub-Saharan Africa, South Asia, and South-East Asia. According to the Centers for Disease Control and Prevention(CDC), lower respiratory diseases are the second leading cause of death in low-income countries, Kenya included. According to the Ministry of Health(MoH), pneumonia is the leading cause of death among children in Kenya, with Kericho, Kisumu, and West Pokot counties bearing the highest burden. Opuba and Onyango (2022) established that pneumonia accounts for 16% of all child deaths in Kenya and is the second killer of children under five. They further stated that the pneumonia burden is attributed to inadequate access to healthcare and poor health-seeking by caregivers. Children can be protected from pneumonia, prevented with simple interventions, and treated with low-cost, low-tech medication and care. However, this can only be done if the disease is diagnosed on time, Hashmi et al. (2020).

According to Dadonaite et al. (2018), 15% of all child deaths in 2017 were caused by pneumonia, confirming that it is the leading cause of death of children. The death rate for old people fell slightly in 2017 compared to the previous years, but the number of deaths of people 70 and older increased. 1.13 million who died from pneumonia were in this age group. Children are most likely to die from pneumonia across Sub-Saharan Africa and South Asia. The deaths in just 5 countries — India, Nigeria, Pakistan, the Democratic Republic of Congo, and Ethiopia – accounted for more than half of all deaths from childhood pneumonia in 2019 Watkins and Sridhar (2018).

American Thoracic Society states that pneumonia is the world's leading cause of death, attributing to 16% of fatalities globally. In 2015, 2400 pneumonia deaths were reported daily and 120 million episodes were reported yearly, amongst which 10% progressed to severe. In 2016 United Nations Children's Fund(UNICEF) reported 880,000 deaths, among which most were infants below 2 years. In the United States, pneumonia is less fatal for children but still a problem in the national hospitals. As half of pneumonia deaths are between 18-64, young people are also affected by pneumonia. This is because pneumonia is the most common cause of sepsis and septic shock. UNICEF also stated that every 43 seconds, a child dies of pneumonia. According to the latest WHO data published in 2020, influenza and Pneumonia deaths in Kenya reached 22,571 or 8.57% of total deaths. The

age-adjusted death rate is 85.68 per 100,000 of the population, ranking Kenya as the 38th leading in the world. 15% of child deaths were due to pneumonia in 2018, and it was the second biggest killer of children under five in 2017. Pneumonia killed almost 9,000 children under five in 2018; more than 1 child every hour.6% is the average annual rate of reduction in pneumonia mortality between 2000–2018. At the same rate, Kenya is expected to reach the 2025 GAPPD target in 2029 with a 4-year gap.

Pneumonia disease is prevalent among young children below 5 years old. According to the report released by "Our World in Data", children below five have the highest death rate caused by pneumonia. To curb and minimize these deaths, there is a need for detection of the disease in its emergence in the early stages when less damage is done, which is easily treatable by antibiotics. Pneumonia is a communicable disease and can be spread in several ways. These include and are not limited to inhaling viruses and bacteria commonly found in the nose or throat into the lungs, through air-borne droplets from a cough or sneeze, and through blood, especially during and shortly after birth.

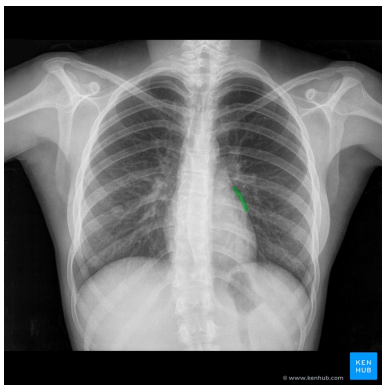
According to the CDC, once infected, the presenting features vary from mild to severe depending on the type of infection, age of the patient, and overall health. In some instances, patients might be infected but present no features, especially children under five. WHO(2021) states that most pneumonia symptoms are mistaken for flu or common cold, and failure to diagnose the disease at its early stages has proven fatal. Medical practitioners have brought forward procedures for diagnosing pneumonia in patients presenting disease symptoms. In the initial step, a doctor examines a patient's medical history and conducts a physical exam. In case pneumonia is suspected, the following tests are recommended:

- Blood tests confirm an infection and identify the type of organism causing the infection.
- Chest X-ray diagnoses pneumonia and determines the extent and location of the infection.
- Pulse oximetry measures the oxygen level in your blood. Pneumonia can prevent your lungs from moving enough oxygen into your bloodstream.
- Sputum test, which involves a sample of fluid from the lungs, is taken after a deep cough and analyzed to help pinpoint the cause of the infection.

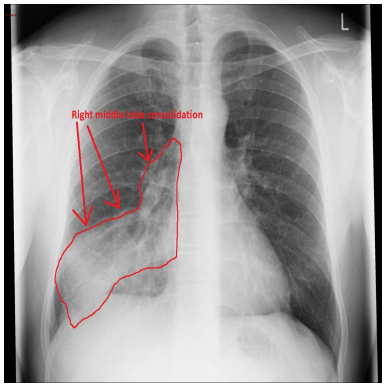
- A chest CT scan can obtain a more detailed image of the lungs.
- Pleural fluid culture where a fluid sample is taken by putting a needle between the ribs from the pleural area and analyzed to help determine the type of infection.

This study mainly focused on Chest X-ray images as an efficient way of diagnosing pneumonia in patients.

X-rays are high-energy radiation waves that can penetrate most substances in varying degrees in low doses to produce images that help doctors diagnose diseases. In high doses, X-rays are used to treat cancer in a procedure known as radiation therapy. X-rays have become increasingly cheaper due to rapid technological advancements such as biomedical equipment. The Chest X-ray is, therefore, the most commonly used procedure in detecting pulmonary diseases like pneumonia. X-ray is preferred over CT scans because CT scans typically take considerably longer than X-ray imaging, and sufficient high-quality CT scanners may not be available in many underdeveloped regions. In contrast, X-rays are the most common and widely available diagnostic imaging technique, crucial in clinical care and epidemiological studies. When an X-ray image is taken, Metal appears completely white; Bone appears almost white; Fat, muscle, and fluids appear shades of grey, while Air and gas appear black. A normal chest X-ray image appears, as shown below.



Medical experts must study the images keenly to arrive at an accurate diagnosis. In the case of Pneumonia diagnosis, the black air in the lungs contrasts clearly with the white infected tissues, which block more of the X-rays. The figure below indicates a pneumonia-infected chest X-ray image.



There are several regions across the globe with scarce availability of practiced healthcare workers, radiologists, and medical imaging experts whose prediction of such diseases matters greatly and therefore creates an urgent need for computerizing the diagnosis. This study, therefore, developed a model that can perform binary classification to detect the presence and absence of pneumonia from patients' X-rays, especially in the early stages of infection. Early and accurate diagnosis is crucial for initiating appropriate treatment and reducing the morbidity and mortality associated with pneumonia.

Chest X-rays have long been a valuable tool in the diagnostic process, allowing healthcare professionals to visualize the lungs and identify characteristic signs of pneumonia, such as infiltrates and consolidations. However, accurate interpretation of chest X-rays for pneumonia diagnosis requires a high level of expertise and experience. Interpretation challenges arise due to the variability in image presentation, overlapping features with other respiratory conditions, and the subjective nature of visual interpretation. These factors can lead to diagnostic errors, delays in treatment initiation, and subsequent negative impacts on patient outcomes.

Artificial intelligence and deep learning models are becoming quite popular in medicine. They are used as problem solvers in diagnoses to provide solutions beyond medical practitioners' knowledge at minimum costs. They act as supplements in making clinical decisions, saving cost and time. Milletari et al. (2016) proposed a methodology for depicting prostrate in MRI volumes using a Convolutional neural network. In 2017, Curigliano et al. (2017) presented a deep learning model for dermatologist-level classification of skin cancer, thus demonstrating an artificial intelligence capable of classifying skin cancer with a level of competence comparable to dermatologists. Other researchers have also used these models to diagnose diseases from X-rays, Nasrullah et al. (2019) used two deep three-dimensional

(3D) customized mixed link network (CMixNet) architectures for lung nodule detection and classification.

Artificial intelligence models have also been used to detect and diagnose pneumonia; Aledhari et al. (2019), Yee and Raymond (2020) and Erdem and AYDİN (2021) used the convolution neural network model for diagnosis. Hashmi et al. (2020) used deep transfer learning with several models, and Ayan et al. (2022) used an ensemble of deep learning models, just to mention a few all to diagnose pneumonia from chest X-rays. Convolution neural network(CNN) models have been used extensively for this purpose and have proven effective, but CNN models are computationally really complex and require very large amounts of data. Ensembles, deep transfer learning, and many other machine learning techniques require high-level expertise as they are also complex, costly, and time-consuming.

Recent advancements in deep learning and neural network models have shown great potential in automating medical image analysis tasks. CapsNet, a novel neural network architecture proposed by Sabour et al., offers a unique approach by introducing capsules, which capture spatial relationships and hierarchical representations within images. Using capsules in CapsNet allows for more robust and accurate feature extraction compared to traditional convolutional neural networks (CNNs), making it suitable for complex image classification tasks. While deep learning models have been applied to pneumonia diagnosis from chest X-rays, there is still room for improvement in accuracy and efficiency. The adoption of CapsNet for pneumonia detection can potentially address the limitations of traditional CNNs and enhance the diagnostic capabilities of automated image classification systems. Therefore, this study leveraged the benefits of CapsNet neural network models to improve the accuracy and efficiency of early pneumonia diagnosis from chest X-rays. By utilizing the spatial relationships and hierarchical representations captured by capsules, the CapsNet model potentially overcame the challenges of chest X-ray interpretation and provided more accurate and timely pneumonia diagnoses.

This research contributed to the field of medical image analysis by providing an innovative approach that can assist healthcare professionals in making accurate and timely pneumonia diagnoses. The development of an automated image classification system based on a CapsNet model has the potential to significantly improve patient outcomes by facilitating early intervention and appropriate treatment strategies.

1.2 Problem Statement

Pneumonia is a significant health threat, ranking as the second leading cause of death in Kenya, with alarming global statistics. Hospital Clínic de Barcelona's research highlights that pneumonia accounted for 2.5 million deaths worldwide in 2019, including 600,000 child fatalities under five years of age. UNICEF reports that pneumonia claims more young lives than any other infectious disease, causing over 700,000 child deaths annually, including newborns. Tragically, these deaths are mostly preventable.

Globally, pneumonia affects more children under five than malaria and diarrhea combined, with over 870,000 children succumbing yearly. This disease represents approximately 15% of all child deaths annually. Alarming statistics from 2019 reveal that 2.5 million people died of pneumonia, with nearly a third being children under five, Dadonaite et al. (2018). This makes pneumonia the leading cause of death in this age group.

Diagnosing pneumonia through chest X-rays is crucial for timely and effective treatment. However, this process faces challenges due to the subjectivity in visual interpretation and image presentation variations, leading to diagnostic errors and delays. Traditional methods like sputum culture are time-consuming, hindering effective care delivery. Therefore, there was an urgent need to develop more accurate and efficient diagnostic measures, especially as deep learning models, while promising, require improvements in their clinical utility.

To address these pressing issues, this study aimed to develop an automated image classification system using a CapsNet neural network model for early pneumonia diagnosis from chest X-rays. The system seeks to enhance accuracy by overcoming challenges like image variability, overlapping features with other respiratory conditions, and ambiguous cases. Simultaneously, it aims to improve efficiency by reducing diagnosis time and resource requirements, all without compromising accuracy. This research sought to significantly enhance the accuracy and efficiency of early pneumonia diagnosis from chest X-rays, enabling timely interventions and ultimately improving patient outcomes.

The urgency of early detection is highlighted by Kallander et al. (2016) with a statement: "We call for action to improve the early identification and treatment of childhood pneumonia at the community and outpatient levels to reduce deaths." The high case-fatality rate in untreated children with pneumonia can reach up to 20%, with fatalities occurring as early

as three days after the onset of illness. This research endeavored to combat pneumonia's devastating impact by revolutionizing its diagnosis, ensuring patients receive the prompt care they urgently need.

1.3 Research questions

1. How can a customized capsule network model be developed and optimized specifically for pneumonia diagnosis from chest X-rays?

2. What is the impact of incorporating capsule-based features in the capsule network compared to traditional CNN-based models in improving the accuracy of pneumonia classification?

3. How can the challenges associated with variability in image presentation and quality, overlapping features in lung structure with other respiratory conditions, and subtle abnormalities be addressed to improve the accuracy of pneumonia classification?

4. How does the developed CapsNet-based system compare to existing approaches in terms of accuracy and efficiency in the early diagnosis of pneumonia from chest X-rays?

1.4 Objectives

1.4.1 Main Objective

To develop an automated image classification system for early diagnosis of pneumonia from chest X-rays using a Capsule network model.

1.4.2 Specific objectives

1. To transform and pre-process chest X-ray image data to capture subtle abnormalities and occlusions.

2. To train a CapsNet model with optimal parameters.

3. To evaluate the performance and accuracy of a capsule model in diagnosing pneumonia from chest X-ray images.

4. To conduct extensive evaluations and comparative analyses to assess the accuracy and efficiency of the developed CapsNet model.

1.5 Justification

Pneumonia affects many individuals, especially children, primarily in developing and underdeveloped countries. It is characterized by risk factors such as overcrowding, poor hygienic conditions, malnutrition, and the unavailability of appropriate medical facilities. Pneumonia is a major cause of death amongst individuals worldwide and a major cause of death for children below five years of age. According to the World Health Organization report in 2022, it accounts for 14% of all deaths of children under five. The symptoms associated with pneumonia usually worsen over a day or two, including an intensifying cough, shortness of breath, and muscle soreness.

Early diagnosis of pneumonia is crucial to cure the disease completely. It helps facilitate prompt and effective treatment so patients benefit from improved outcomes. Most people recover in 2 to 4 weeks, but babies, the elderly, those with lung or heart issues, and those who are pregnant risk becoming extremely unwell and may require hospitalization. The high incidence of severe pneumonia in many hospitals can be explained by delayed care-seeking behavior, limited hospital care access, or both. There was, therefore dire need to diagnose the disease at its onset to ensure treatment and lower fatalities. Without early intervention, pneumonia progresses to a state where even intravenous antibiotics have limited impact, leading to high hospital case–fatality rates, especially for children Kinyoki et al. (2017).

Early diagnosis of pneumonia significantly impacts health globally since pneumonia is a leading cause of morbidity and mortality worldwide, particularly among vulnerable populations. Timely and accurate diagnosis is critical for initiating appropriate treatment and improving patient outcomes. By developing an automated image classification system, this research significantly impacted global health by facilitating early intervention and improving patient care. In addition to advancement in global health, this study also solved diagnostic challenges experienced during manual interpretation of X-rays. Chest X-ray di-

agnosis of pneumonia is a challenging undertaking that calls for knowledge and experience. However, it is prone to interpreting mistakes, which can delay diagnoses and produce worse-than-ideal patient outcomes. This study intended to solve the drawbacks of conventional methods and enhance the precision of automated pneumonia diagnosis using CapsNet, a distinctive neural network architecture.

This study contributed to the advancement in deep learning and automatic image classification. The CapsNet model, with its ability to capture spatial relationships and hierarchical representations, offers a unique approach to image classification. This research contributes to the field by exploring the application of CapsNet in pneumonia diagnosis and comparing its performance to existing approaches. This will ensure healthcare delivery efficiency as manual interpretation of chest X-rays for pneumonia diagnosis is time-consuming and resource-intensive. By developing an efficient automated image classification system, this research potentially reduces the burden on healthcare professionals, accelerates the diagnostic process, and optimizes resource allocation. This can lead to more effective utilization of healthcare resources and improved efficiency in healthcare delivery.

The outcomes of this research have practical implications for clinical settings. The developed CapsNet-based system can be integrated into existing healthcare infrastructure, providing a reliable tool for healthcare professionals to support their diagnostic decision-making process. The system's accuracy and efficiency improvements have the potential to impact patient outcomes and contribute to better healthcare practices positively.

1.6 Significance of the study

This study was mainly aimed at developing a model for the early diagnosis of pneumonia, which is significant in several ways:

- Improved patient outcomes: Pneumonia is a major cause of morbidity and mortality worldwide. Timely and accurate diagnosis is crucial in initiating appropriate treatment interventions, leading to improved patient outcomes. This study's outcomes will assist healthcare professionals in making more precise and timely diagnoses, resulting in better patient care and potentially saving lives.

- Reduced diagnostic errors and delays: Manual interpretation of chest X-rays for pneu-

monia diagnosis is subjective and prone to errors, leading to diagnostic delays and suboptimal treatment decisions. Using a CapsNet neural network model offers the potential to overcome these limitations by providing a standardized and automated approach to image classification. This research contributes to more efficient and effective healthcare delivery by reducing diagnostic errors and delays.

- Enhanced efficiency in healthcare: Implementing an automated image classification system based on CapsNet can significantly improve the efficiency of pneumonia diagnosis. By reducing the reliance on manual interpretation, healthcare professionals can allocate their time and resources more effectively, leading to faster diagnoses and streamlined workflows. This can positively impact healthcare systems by optimizing resource utilization and improving patient throughput.

- Advancements in medical imaging technology: The study's focus on utilizing CapsNet, a novel neural network architecture for pneumonia diagnosis from chest X-rays, contributes to the advancement of medical imaging technology. By exploring the capabilities of CapsNet in capturing spatial relationships and hierarchical representations within images, this research expands the knowledge base of deep learning models in medical image analysis. The findings paved the way for further advancements in automated image classification systems for various medical conditions.

- Potential for widespread adoption: The development of an accurate and efficient automated image classification system for pneumonia diagnosis has the potential for widespread adoption in clinical settings. Integrating such a system into existing healthcare infrastructure can provide healthcare professionals a reliable tool to support their decision-making process. This can lead to improved diagnosis standardization, reduced interpretation variability, and improved overall quality of care.

By addressing these significant aspects, this study potentially advanced the field of pneumonia diagnosis, ensuring improved patient care, and paved the way for further research and development in automated medical image analysis.

1.7 Limitations of the study

- Dataset limitations: The availability of a comprehensive and well-annotated dataset is crucial for training and evaluating the CapsNet model. However, obtaining a sufficiently large and diverse dataset of chest X-ray images with accurate pneumonia annotations can be challenging.

- Generalizability to diverse populations: Medical conditions, including pneumonia, can vary across populations and demographics. The dataset used for training the CapsNet model may primarily represent a specific population or demographic group, which can limit the model's generalizability to other populations. This may affect the accuracy and effectiveness of the model when applied to different patient cohorts.

- Limited clinical validation: The study focused on developing an automated image classification system for pneumonia diagnosis. However, limited clinical validation may be conducted due to various practical constraints, such as time, resources, and access to patient data. This may hinder the ability to assess the real-world performance and impact of the developed system on clinical outcomes.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

This section explored and provided an overview of existing research and studies relevant to the early diagnosis of pneumonia from chest X-rays using deep learning models. It explored the field's current state, identified key research gaps, and established the foundation for the study.

2.2 Pneumonia diagnosis and chest x-rays

Pneumonia, a common respiratory infection, poses a significant health burden globally. It affects millions yearly, leading to substantial morbidity and mortality (WHO, 2019). Timely and accurately diagnosing pneumonia is crucial for appropriate patient management, effective treatment, and preventing complications. Among the diagnostic tools available, chest X-rays have been widely used as a primary imaging modality for pneumonia detection (UNICEF, 2019b). Chest X-rays provide valuable insights into the pulmonary structures and are often the first-line imaging technique used to assess patients with respiratory symptoms (CDC). These radiographic images allow healthcare professionals to visualize abnormalities indicative of pneumonia, such as infiltrates, consolidations, and opacities (UNICEF, 2019b). However, the interpretation of chest X-rays heavily relies on the expertise and experience of radiologists and clinicians, making it subjective and prone to inter-observer variability (UNICEF, 2019b).

Manual interpretation of chest X-rays for pneumonia diagnosis has inherent limitations. Radiologists may experience difficulties in distinguishing between pneumonia and other lung conditions with similar radiographic findings. Moreover, the process can be time-consuming, leading to diagnosis and treatment initiation delays. The growing demand for accurate and efficient pneumonia diagnosis has driven researchers to explore automated image classification techniques, particularly leveraging the power of deep learning.

Deep learning, a subset of artificial intelligence, has revolutionized various fields, including computer vision and image analysis. Automated image classification models built on deep learning techniques offer several potential advantages for pneumonia diagnosis. They can quickly process large volumes of data, detect subtle abnormalities, and provide consistent and objective assessments. Furthermore, these models can potentially improve the efficiency of pneumonia diagnosis, allowing healthcare professionals to prioritize and expedite treatment plans for affected individuals. By leveraging the power of deep learning and automated image classification, we can potentially enhance the accuracy, efficiency, and accessibility of pneumonia diagnosis from chest X-ray images. This can significantly impact patient outcomes, clinical decision-making, and public health strategies.

2.3 Traditional methods

Pneumonia diagnosis has traditionally relied on manual interpretation of chest X-rays by experienced radiologists and clinicians. These healthcare professionals visually inspect the radiographic images for abnormalities indicative of pneumonia, such as infiltrates, consolidations, and opacities in the lung fields. While this approach has been widely used, it has several limitations that can impact accuracy, efficiency, and inter-observer consistency. The primary challenge with manual interpretation is the subjective nature of visual assessment. Radiologists' interpretations can vary due to differences in experience, expertise, and individual biases. This subjectivity can lead to inconsistencies and discrepancies in diagnosis, affecting patient care and treatment decisions. Moreover, the reliance on human visual inspection introduces the possibility of errors and oversights, mainly when dealing with subtle or complex cases.

Manual interpretation of chest X-rays is also a time-consuming process. Radiologists

need to review many images, often leading to backlogs and delays in diagnosis. This delay can adversely affect patient outcomes, as timely initiating appropriate treatment is crucial for managing pneumonia effectively and preventing complications. Furthermore, the increasing demand for radiological services can strain healthcare systems, exacerbating time constraints and adding pressure on radiologists to deliver rapid and accurate diagnoses. Another limitation of traditional approaches is distinguishing pneumonia from other lung conditions that may present with similar radiographic findings. Conditions such as lung abscesses, atelectasis, or even certain tumors can exhibit radiographic features that overlap with pneumonia. Accurate differentiation is critical to ensure appropriate treatment strategies and avoid unnecessary interventions.

Researchers have explored the potential of automated image classification techniques to address these limitations and improve pneumonia diagnosis. By leveraging advancements in machine learning and deep learning, automated models can assist in the analysis of chest X-ray images, provide objective assessments, and aid in diagnosing and classifying pneumonia.

2.4 Deep learning and Image Classification

Deep learning has emerged as a powerful subset of machine learning that has revolutionized various fields, including computer vision and image analysis. It has shown remarkable success in solving complex problems by automatically learning hierarchical representations of data, particularly in the form of artificial neural networks. Deep learning models have become the go-to approach for image classification tasks; for instance, CNNs are designed to mimic the human visual system's hierarchical organization, allowing them to learn and extract meaningful features from images effectively. The key advantage of deep learning lies in its ability to automatically learn intricate patterns and representations from large amounts of data. Unlike traditional machine learning methods that rely on manual feature engineering, deep learning models can autonomously learn the features directly from the raw data. This makes them highly effective in capturing complex and abstract patterns that may be difficult to define explicitly.

Deep learning techniques have demonstrated great promise in the context of pneumonia

diagnosis from chest X-ray images. By training CNN models on large-scale datasets, such as the ChestX-ray14 dataset, these models can learn to distinguish regular chest X-rays from those exhibiting pneumonia-related abnormalities. They can automatically learn the subtle patterns and characteristics indicative of pneumonia, allowing for accurate classification. Image classification is a fundamental task within deep learning, where the goal is to assign a label or class to an input image based on its content. In the case of pneumonia diagnosis, image classification models can learn to differentiate between normal and abnormal chest X-rays, enabling the detection and identification of pneumonia-related features.

The process of image classification involves two main steps: training and inference. During the training phase, the deep learning model learns the underlying patterns and relationships in the training dataset through an iterative optimization process. It adjusts the parameters of the neural network to minimize the discrepancy between the predicted labels and the true labels of the training samples. Once trained, the model can infer new, unseen chest X-ray images. It applies the learned knowledge to classify the images as normal or indicative of pneumonia. The model's predictions are based on the patterns and features learned during the training phase.

Deep learning models, particularly CNNs, have performed remarkably in various image classification tasks, including pneumonia diagnosis from chest X-ray images. They have achieved high accuracy rates, often comparable to or surpassing human radiologists' performance. This suggests their potential as valuable tools in assisting healthcare professionals in diagnosing pneumonia and improving efficiency, accuracy, and patient outcomes. In recent years, many researchers have highlighted the importance of the prompt detection and diagnosis of pneumonia and other diseases using deep learning techniques. Gulshan et al. (2016) focused on developing and validating a deep-learning algorithm for detecting diabetic retinopathy in retinal fundus photographs. While not specific to pneumonia diagnosis, their work showcased the success of deep learning in medical image analysis. Lakhani and Sundaram (2017) utilized convolutional neural networks for the automated classification of pulmonary tuberculosis on chest radiographs. While their study specifically targeted tuberculosis, the application of deep learning in pneumonia diagnosis shares similarities regarding image analysis and classification.

In 2017, Rajpurkar et al. (2017) developed CheXNet, a deep-learning model capable of

achieving radiologist-level accuracy in pneumonia detection on chest X-rays. Their study demonstrated the potential of deep learning algorithms to aid in diagnosing pneumonia from X-ray images. Wang et al. (2017) introduced the ChestX-ray8 database, a large-scale dataset for weakly-supervised classification and localization of common thorax diseases, including pneumonia. Later in 2019, Irvin et al. (2019) introduced CheXpert, a large dataset of chest radiographs with expert comparison and uncertainty labels. This dataset provides valuable resources for training and evaluating deep-learning models for pneumonia diagnosis and other chest-related conditions.

Lee et al. (2019) developed a deep learning-based system to improve the detection of pulmonary nodules on chest radiographs, demonstrating the potential of deep learning in enhancing pneumonia diagnosis accuracy. Furthermore, Chassagnon et al. (2020) provided insights into recent advances in the automated detection of pulmonary nodules on chest CT scans, which can inform the development of advanced diagnostic systems for pneumonia. A survey by Kareem et al. (2022) reviews pneumonia and image detection by examining how different machine learning technologies like convolution neural network (CNN), k-nearest neighbor (KNN), RESNET, CheXNet, DECNET, and artificial neural network (ANN) can be used in detecting pneumonia disease. They reviewed the literature comprehensively to find how we can combine hospitals and medical institutions to train machine learning models from their datasets so that the ML algorithms can detect diseases more efficiently and correctly.

These studies collectively contribute to the literature on pneumonia diagnosis and highlight the potential of deep learning and machine learning in improving the accuracy and efficiency of pneumonia detection from chest X-ray images.

2.5 Advances in Pneumonia Diagnosis

In recent years, deep learning techniques, particularly convolutional neural networks (CNNs), have significantly advanced pneumonia diagnosis using chest X-ray images. These approaches leverage the power of deep learning to automatically learn and extract meaningful features from radiographic images, leading to improved accuracy and efficiency in pneumonia detection. One of the notable advancements in pneumonia diagnosis is the de-

velopment of large-scale datasets specifically curated for training deep-learning models. The ChestX-ray14 dataset, consisting of over 100,000 chest X-ray images, has become a benchmark for pneumonia diagnosis research. This dataset provides a diverse range of cases, including both pneumonia and non-pneumonia samples, enabling deep-learning models to learn discriminative features for accurate classification. Deep learning models trained on these datasets have achieved remarkable performance in pneumonia diagnosis. They have demonstrated high accuracy in distinguishing between normal and abnormal chest X-rays, including those indicating pneumonia. Studies have reported sensitivity and specificity levels comparable to or surpass human radiologists' performance, indicating the potential of deep learning models as valuable tools for pneumonia diagnosis.

Convolution Neural network, in particular, has been widely used in computer vision and problems involving very large complex data sets that do not promise any clear statistical distributions.Gupta (2021) developed a CNN model for detecting pneumonia from chest x-ray images and achieved validation accuracy of 93% and training accuracy of about 99%.GM et al. (2021) performed a study on the detection of pneumonia using 15 different CNN models to find a model that is easy to train and computationally less expensive. They concluded that there is a need for extensive studies to obtain simple models that can maintain high-level accuracy and performance in diagnosing pneumonia. It is therefore evident that CNN models are limited in that they require very high computation power and can take a lot of time during training hence time inefficient. CNN also requires large amounts of datasets to process and train the model.

Ensemble learning has also been investigated to enhance the robustness and accuracy of pneumonia diagnosis models. By combining predictions from multiple deep learning models, ensemble models can capture a diverse range of features and improve the overall classification performance. Ensemble methods have shown promising results, further increasing the accuracy and reliability of pneumonia diagnosis. Sirazitdinov et al. (2019) proposed an ensemble model based on two CNN models, RetinaNet and mask R-CNN, for pneumonia detection and localization from a large-scale chest x-ray database. They presented the superiority of focal loss and object detection in classification metrics, but accuracy levels indicated a need for extensions in the study.

Liz et al. (2021), used ensembles of CNN models to diagnose paediatric pneumonia to

maximize the Area Under the curve (AUC) and the true positivity rate TPR. Their study was mainly focused on addressing the challenges in CNN models with highly unexplainable outputs that make the classification process rigorous, as well as the fact that they require very large datasets for training to achieve a high degree of accuracy. The ensemble model developed was a new explainable artificial intelligence technique based on combining the individual heatmaps obtained from each model in the ensemble, and it was able to achieve robust results compared to Kermany et al. (2018b) study, which used deep transfer learning to distinguish viral from bacterial pneumonia from chest x-rays.

Ayan et al. (2022) used an ensemble of deep convolutional neural networks to diagnose paediatric Pneumonia in Chest X-ray images. They used seven CNN models VGG-16, VGG-19, ResNet-50, Inception-V3, Xception, MobileNet, and SqueezeNet, pre-trained on the ImageNet dataset and trained with the appropriate transfer learning and fine-tuning strategies on the chest X-ray dataset. They achieved a classification accuracy of 90.71 in chest X-ray images as normal, viral, and bacterial pneumonia. According to Ayan et al. (2022), determining hyperparameters of pre-trained CNN models while using transfer learning and fine-tuning reveals a long trial-and-error process. At the same time, high variance and low bias are important dual to ensemble CNN models. Ensembled learning models are limited because they sometimes lack interpretability and require large datasets and high computational power. The models are very complex and require a high level of know-how to develop. Furthermore, they are computationally expensive, adding learning time and memory constraints to the problem.

Researchers have also explored transfer learning, where pre-trained CNN models on large-scale image datasets, such as ImageNet, are fine-tuned on pneumonia-specific datasets. Transfer learning allows models to leverage knowledge learned from vast amounts of data, improving generalization and performance on pneumonia detection tasks. Kermany et al. (2018b) used transfer learning to diagnose paediatric pneumonia. The model used was highly accurate in diagnosing pneumonia as well as distinguishing between Viral and Bacterial pneumonia but was limited in that training a highly accurate model with a small data set will provide inferior performance to that of a model trained from a random initialization on an extremely large data-set which would take weeks to achieve accuracy. This indicates that transfer learning requires large datasets for training to achieve high accuracy, requiring

more time.

Liang and Zheng (2020) proposed a residual structure and dilated convolution to classify paediatric pneumonia images and used transfer learning to initialize model weight parameters. The study's main aim was to solve the problem of low image resolution and overlap in the inflamed area in the chest X-ray. The model achieved a classification accuracy of 96.7% and an f1-score of 92.7%. A recent study by Rehman et al. (2022) to develop an efficient diagnosis of COVID-19 disease by differentiating it from viral pneumonia, bacterial pneumonia, and healthy cases used transfer learning techniques to pre-train the model. Despite achieving accuracy in the results, many CT and X-ray datasets were required. Transfer learning also is limited in that it requires large data sets to pre-train the model to achieve perfect weights and accuracy of the model. Transfer learning is inefficient as a model can take weeks to learn images if the data is very large. Additionally, transfer learning architectures are broad and many, and choosing a perfect architecture is key to obtaining a perfect diagnosis in case of diseases.

Integrating deep learning models with clinical decision support systems (CDSS) has also been explored to enhance pneumonia diagnosis. CDSS platforms leverage deep learning models' capabilities to analyze chest X-ray images and provide real-time diagnostic recommendations to healthcare professionals. These systems aim to assist radiologists and clinicians in making accurate and timely diagnoses, potentially improving patient outcomes. While these advances in deep learning for pneumonia diagnosis are promising, several challenges and considerations remain. The lack of time and computationally efficient models to be used even when resources are limited and for small datasets needs to be addressed. Additionally, the interpretability and explainability of deep learning models in pneumonia diagnosis are areas that require further research to build trust and facilitate integration into clinical practice. Future research should address the remaining challenges to fully harness the potential of deep learning in pneumonia diagnosis and enhance patient care.

2.6 Capsule Networks for Pneumonia Diagnosis

In recent years, there has been growing interest in exploring the application of capsule networks (CapsNets) for pneumonia diagnosis from chest X-ray images. CapsNets, an

alternative architecture to convolutional neural networks (CNNs), aim to overcome some of the limitations of traditional CNNs by capturing richer spatial relationships and hierarchical representations of objects in images. Sabour et al. (2017) proposed the concept of CapsNet and dynamic routing for enhancing deep learning architectures; their work introduced the idea of capsules and routing mechanisms that can capture hierarchical relationships in image analysis tasks.

The key concept behind CapsNets is "capsules," which are groups of neurons that represent specific object features and their instantiation parameters. Unlike CNNs, which rely on individual activation maps to detect specific features, capsules aim to encode information about objects' presence, pose, and relationship within an image. This makes CapsNets potentially more robust and interpretable, as they can capture the relative positions, orientations, and sizes of different features. These capsules capture features' presence and spatial relationships, allowing for more robust and interpretable representations. CapsNets have shown promise in improving the interpretability of deep learning models, which can aid in understanding the underlying factors contributing to pneumonia diagnosis.

Yan et al. (2020) investigated the use of capsule networks for lung nodule classification using multi-view data. Although their focus was on lung nodules, utilizing capsule networks highlights the potential for capturing complex patterns and improving classification accuracy in medical imaging tasks. In the context of pneumonia diagnosis, CapsNets offer several advantages. By encoding spatial relationships between abnormalities in the lung, CapsNets may provide a more holistic representation of pneumonia-related features. This can aid in capturing subtle patterns and variations that traditional CNNs might miss. Furthermore, the hierarchical nature of CapsNets allows for the modeling of complex interactions between different abnormalities and their combinations, potentially leading to more accurate and reliable diagnoses.

Capsule Networks (CapsNets) present a promising approach for pneumonia diagnosis from chest X-ray images. Their ability to capture spatial relationships and hierarchical representations of abnormalities offers potential advantages over traditional CNNs. While the field is still emerging, early studies indicate that CapsNets can potentially improve the accuracy and interpretability of pneumonia diagnosis. Further research is needed to explore the full capabilities of CapsNets, optimize their performance, and validate their efficacy in

clinical settings.

2.7 Conclusion

The application of deep learning techniques has shown remarkable advancements in pneumonia diagnosis from chest X-ray images. These techniques have demonstrated the potential to enhance the accuracy, efficiency, and reliability of automated image classification systems for pneumonia detection. By training deep learning models on large-scale datasets, such as the ChestX-ray14 dataset, these models can learn to differentiate between normal and abnormal chest X-rays, including those indicative of pneumonia-related abnormalities. They can capture intricate patterns and subtle variations that may be challenging to identify through manual analysis.

The utilization of deep learning techniques, particularly CapsNets, can revolutionize pneumonia diagnosis from chest X-ray images. These techniques offer improved accuracy, efficiency, and clinical utility, thereby assisting healthcare professionals in making timely and accurate diagnoses. By addressing the remaining challenges and pursuing further research, we can harness the full potential of deep learning in pneumonia diagnosis, ultimately enhancing patient care and outcomes.

Chapter 3

METHODOLOGY

3.1 Introduction

The methodology entails all the procedures used to achieve this research's objectives scientifically and validly. It includes the research design, data, data manipulation, and analysis procedures, all organized to achieve the goal of this study.

3.2 Data source

Chest X-rays play a crucial role in diagnosing diseases across hospitals worldwide daily. However, obtaining organized and relevant raw data for image classification can be challenging. This study used secondary data from Mendeley data. The dataset was developed and uploaded by Kermany et al. (2018a) in 2018. The dataset contains thousands of validated OCT and Chest X-ray images described and analyzed in "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning." The dataset is readily available for research purposes. By leveraging this existing dataset, the goal was to explore and improve the classification of chest X-ray images, contributing to advancing medical diagnostics and potentially benefiting patients globally.

3.3 Research design

This study employed a cross-sectional research design, a cross-sectional research design is a type of study that aims to collect data from a sample population at a single point

in time. It involves gathering information about a specific phenomenon interest from a representative sample to analyze the relationship between different variables. The chest X-ray images were collected once from each patient to form a representative sample.

3.4 Data analysis

Data analysis entails inspecting, cleaning, transforming, and interpreting raw data to uncover patterns, discover insights, and make informed decisions. It involves using various statistical and analytical techniques to extract meaning from data and derive valuable information. This study used the following step-wise procedure for data manipulation and analysis:

Python programming was used to perform all these analysis procedures. Python is a high-level open-source general-purpose programming language. It is a popular and widely used language in data analysis due to its key advantages. Python has a simple and intuitive syntax, making it accessible to beginner and experienced programmers. It allows researchers to focus more on the data analysis rather than getting caught up in complex programming details. It also offers a rich ecosystem of libraries specifically designed for data analysis, such as NumPy, Pandas, and Matplotlib. These libraries provide powerful tools and functions for data manipulation, analysis, visualization, and statistical modeling.

Python seamlessly integrates with popular machine learning libraries like Scikit-learn and TensorFlow, enabling researchers to build and evaluate complex models for classification, regression, and other tasks. This is particularly valuable in image classification, where machine learning algorithms are commonly employed. Python promotes reproducibility in research by enabling the creation of clean, well-documented code that can be easily shared and replicated. Its ability to efficiently handle large datasets makes it suitable for scalable data analysis tasks, ensuring researchers can effectively tackle substantial volumes of data and derive meaningful insights.

3.5 Data pre-processing

Data pre-processing is a major step in machine learning models as it enhances the model's accuracy. Pre-processing of data cleans the data, fills up the missing values, removes outliers, and identifies specific values in the data. The standard format for radiology images is DICOM-Digital Imaging and Communication in Medicine format for medical images. The DICOM image contains pixel and patient information; we need to convert it to a PNG or JPG format to visualize the image. The Pydicom package extracts a pixel array from a DICOM image, producing a grayscale chest X-ray image. The images in the dataset were provided as grayscale PNG images, and the following pre-processing procedures were carried out.

3.5.1 Image resizing

Image resizing is the process of altering the dimensions of an image while preserving its aspect ratio. It involves adjusting the number of pixels in an image to make it larger or smaller. The chest X-ray images were resized to a standardized resolution. This step ensured uniformity and consistency in the input size of the images, which is crucial for model training.

3.5.2 Image normalization

Image normalization is a technique used to standardize the pixel values of an image by bringing the pixel values within a certain range or distribution, making them more suitable for further analysis or processing tasks. Normalizing an image can improve the performance of machine learning models, enhance image quality, and ensure fair comparisons between different images. This step helps reduce the influence of variations in image brightness and contrast, making the data more suitable for training the model. There are two major ways of normalizing:

- Min-Max normalization (Scaling)- the pixel values are linearly transformed to a new range, typically between 0 and 1. This process involves finding the image's minimum and maximum pixel values and rescaling all the pixel values based on this range. The

formula used is:

$$Normalized.value = \frac{Pixel.value - Min.value}{Max.value - Min.value}$$

- Z-score normalization (Standardization)-the pixel values are transformed to have a mean of 0 and a standard deviation of 1. This is achieved by subtracting the mean value of the image from each pixel value and dividing it by the standard deviation. The formula is:

$$Normalized.value = \frac{Pixel.value - Mean.value}{Standard.deviation}$$

Normalizing the pixel values helps the model converge faster during training, ensures numerical stability, and reduces the impact of different image intensities or lighting conditions on the analysis results.

3.5.3 Image Augmentation

Image augmentation is a technique commonly used in computer vision and deep learning tasks to artificially increase the size and diversity of a training dataset by applying various transformations to the existing images. It is particularly useful when the available dataset is limited, as it helps mitigate overfitting and improves the generalization capabilities of machine learning models. Image augmentation aims to create new training examples visually similar to the original images but exhibit variations in position, scale, rotation, brightness, and other factors. Techniques used in image augmentation include:

1. Flipping: Mirroring an image horizontally or vertically to create a new training example with a different orientation.

2. Rotation: Rotating an image by a certain angle introduces the object's position or orientation variability.

3. Scaling and cropping: Resizing or cropping an image to change its size and aspect ratio.

4. Translation: Shifting an image horizontally or vertically to simulate the object's movement or change in position.

5. Brightness and contrast adjustment: Modifying the image's intensity values to simulate different lighting conditions.

6. Noise addition: Introducing random noise to the image to make it more robust to noise in real-world scenarios.

7. Elastic deformations: Applying local distortions to the image to simulate deformations or variations in shape.

Applying these transformations to the training images, the augmented dataset becomes more extensive and diverse, allowing the model to learn from a broader range of variations and improve its ability to generalize to new, unseen examples. This can lead to better model performance, increased accuracy, and improved robustness.

3.5.4 Data Balancing

Data balancing refers to the techniques used to address the issue of imbalanced class distribution in a dataset. Imbalanced data occurs when one class or category in the dataset is significantly more prevalent than others, leading to skewed learning patterns and potentially biased model performance. Imbalanced data poses challenges for machine learning algorithms, as they favor the majority class and exhibit poor performance on minority classes. To address this issue, various data balancing techniques can be applied. These include:

- Undersampling: This involves randomly reducing the number of instances in the majority class to match the number of instances in the minority class.

- Oversampling: Oversampling involves replicating or creating new instances in the minority class to match the number of instances in the majority class. This can be done through techniques like random duplication, synthetic sample generation, e.g., SMOTE - Synthetic Minority Over-sampling Technique, or other methods that preserve the characteristics of the minority class.

- Class weighting: Class weighting assigns higher weights to instances of the minority class and lower weights to instances of the majority class during model training. This helps the model to pay more attention to the minority class and achieve a balanced decision boundary.

- Ensemble methods: Ensemble methods, such as bagging or boosting, can be used to create multiple models trained on different balanced subsets of the data. The predictions from these models can be combined to obtain a more balanced and accurate final prediction.

3.5.5 Data partitioning

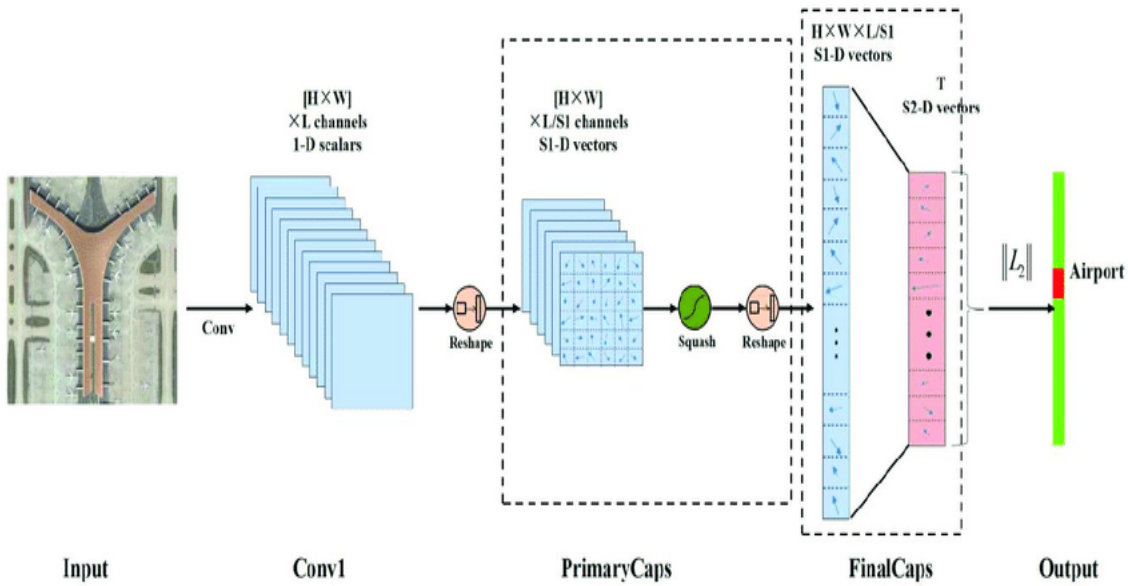
Data partitioning is the process of dividing a dataset into training, validation, and testing subsets. This division helps in avoiding overfitting, where a model becomes too specific to the training data and fails to generalize well to new data. The training set is the largest subset used to train the model by exposing it to various examples. The model learns from the patterns and relationships in this data to make classifications. The validation set is crucial for fine-tuning the model's hyperparameters, such as learning rates or regularization techniques. By evaluating the model's performance on the validation set, adjustments can be made to optimize its performance. Finally, the testing set provides an unbiased evaluation of the trained model's performance. It is a benchmark to measure the model's ability to generalize to new, unseen data, providing insights into its real-world performance.

Data partitioning allows for rigorous evaluation of models and helps ensure they are robust and reliable when deployed in practical applications. It enables the selection of appropriate models and hyperparameters, leading to improved performance and reliable predictions of unseen data.

3.6 The Capsule Network model

The Capsule Network model, also known as CapsNet, is a type of neural network architecture introduced by Geoffrey Hinton and his colleagues in 2017. It was designed as an

alternative to traditional convolutional neural networks (CNNs) and aimed to address some of their limitations, particularly in tasks involving object recognition and understanding spatial relationships. The Capsule Network model is based on the concept of "capsules," which are groups of neurons that encode the instantiation parameters of an entity (such as position, pose, and size) and the presence of that entity in an image. These capsules capture the entity's presence and properties, making them more powerful than individual neurons in CNNs. The proposed capsule model is given as follows:



The key idea behind CapsNet is introducing a dynamic routing mechanism called "routing-by-agreement." This mechanism allows capsules at one layer to communicate with capsules at the next layer, enabling them to reach a consensus on the existence and properties of entities in the input. This routing process considers the agreement between lower-level and higher-level capsules to determine the weights of connections, which facilitates the hierarchical representation of complex spatial relationships. The CapsNet model has shown promising results in various tasks, including digit recognition, object detection, and image segmentation. It has demonstrated better resistance to occlusion and viewpoint changes than traditional CNNs. However, due to its relative novelty, CapsNet is still an active research area, and further advancements and improvements are being explored.

3.6.1 The CapsNet architecture

The CapsNet architecture consists of several layers of capsules that capture entities' hierarchical relationships and properties within input data. The overview of the typical

structure of a CapsNet:

1. Input layer: The input layer receives the raw data, usually images or feature vectors.

2. Primary capsule layer: The primary capsule layer extracts low-level features from the input data.

3. Dense capsule layer: The primary capsule layer is followed by one or more convolutional capsule layers. Each convolutional capsule represents a higher-level feature and is connected to primary capsules through weighted connections. The primary capsules' outputs are transformed into vectors, representing the instantiation parameters (e.g., position, pose), and are then weighted and combined to form the output of the convolutional capsules.

4. Routing-by-agreement mechanism: It involves iterative routing between lower-level and higher-level capsules to establish agreement on the existence and properties of entities. The routing process determines the weights of the connections between capsules based on the agreement, allowing for the representation of hierarchical relationships and spatial configurations.

5. Output layer: The final layer produces the model's classifications. It often consists of capsules that represent specific classes or categories.

Input layer

The input layer is the initial layer that receives the raw input data. It serves as the entry point for information into the network and provides the basis for subsequent layers to process and learn from. In this study, the input layer receives input data as images or feature vectors. A set of primary capsules represents each input; these initial units capture low-level features about specific aspects of the input image. The input layer may consist of multiple nodes, each corresponding to a feature in the image. The values of these neurons represent the intensity of characteristics of the corresponding pixels.

The input layer plays a crucial role in capturing and encoding the initial information from the input data, which is then processed and transformed by subsequent layers in the

network. The output from the input layer is passed to the following layer, the primary capsule layer, where higher-level features and relationships are extracted.

Primary Capsule layer

The primary capsule layer follows the input layer and extracts low-level features from the input data. The input is a set of primary capsules, which are groups of neurons organized in a grid-like structure. Each primary capsule represents a specific local pattern or feature detected in the input data. The primary capsules use convolutional layers to extract low-level features from the input images. Each primary capsule within a group captures variations and properties of a particular feature. These capsules encode not only the presence or absence of the feature but also additional information, such as its pose, position, or other instantiation parameters.

The primary capsules in this layer are connected to the output of the input layer through weighted connections. These connections determine the strength of each primary capsule's output with respect to the input data. The weights of these connections can be learned during the training process, allowing the primary capsules to adapt and specialize in capturing different features. The output of the primary capsule layer is a set of vectors, where each vector represents the instantiation parameters of a specific feature or entity. These vectors carry information about the detected features' presence, properties, and spatial relationships.

The primary capsule's role is to extract meaningful low-level features from the input data and provide a basis for higher-level capsules to capture more complex relationships and interactions between features. By organizing features into capsules, the primary capsule layer enables the network to encode the presence and properties of entities in a more structured and informative manner.

Dense capsule Layer

The primary capsules are connected to the next layer, called the capsule layer. This layer receives the outputs from the primary capsules and captures higher-level features and relationships. The primary capsules provide the low-level feature representations, and the dense capsule layer takes these representations as input. Each capsule in this layer is

connected to all capsules in the previous layer, allowing for communication and agreement among the capsules. The routing-by-agreement mechanism is applied between the primary capsules and the fully connected capsule layer.

The dense capsule layer captures more complex relationships and interactions between features, building upon the low-level features extracted by the primary capsules. The layer's capsules encode higher-level representations of entities, incorporating information about their spatial relationships, poses, and other properties. Each capsule in the capsule layer represents a higher-level concept or feature that encodes the lower-level feature's instantiation parameters (pose).In contrast to traditional CNNs, where the activation of a neuron represents the presence of a feature, in CapsNet, the activation of a capsule represents the probability of the feature's existence and its instantiation parameters.

Output layer

The output layer is the final layer of the network responsible for producing the model's classifications. It transforms the outputs from the preceding dense capsule layers into meaningful probabilities for different classes or categories. In the output layer, each capsule represents a specific class or category. The capsule's output vector encodes information about the class's properties and presence in the input data. The length or magnitude of the output vector represents the probability or confidence associated with that class.

The outputs of the capsules in the output layer are often normalized using a softmax function. The softmax function ensures that the outputs across all classes sum up to 1, allowing them to be interpreted as probabilities. This normalization facilitates interpreting and comparing the model's predictions across different classes.

3.6.2 Routing by Agreement mechanism

The CapsNet architecture employs a routing algorithm to establish dynamic connections between capsules in adjacent layers. The routing-by-agreement mechanism is a critical component of the CapsNet architecture that facilitates communication and consensus among capsules in different layers. It enables the capsules to collaboratively determine the instantiation parameters and existence of entities within the input data. The routing-by-agreement mechanism involves iterative steps to establish agreement or "routing" between lower-level

and higher-level capsules. The overview of how the mechanism works is as follows:

1. Input from lower-level capsules: The lower-level capsules produce output vectors representing the instantiation parameters and properties of entities detected in the input data.

2. Prediction vectors: These output vectors from the lower-level capsules are then transformed into prediction vectors. Each prediction vector represents the likelihood or presence of a specific entity or feature within the input data.

3. Coupling coefficients: Initially, the coupling coefficients, representing the weights of the connections between the lower and higher-level capsules, are set to small values. These coefficients determine how much influence each lower-level capsule has on each higher-level capsule.

4. Iterative routing: The routing process involves iterative steps. The coupling coefficients are adjusted in each iteration based on the agreement between the lower-level capsules' prediction vectors and the higher-level capsules' predictions. The adjustment is made to increase the weights of connections between capsules that exhibit agreement and decrease the weights for capsules with disagreement.

5. Recalculating prediction vectors: After adjusting the coupling coefficients, the prediction vectors from the lower-level capsules are recalculated using the updated weights. This allows the lower-level capsules to refine their predictions based on the feedback from the higher-level capsules.

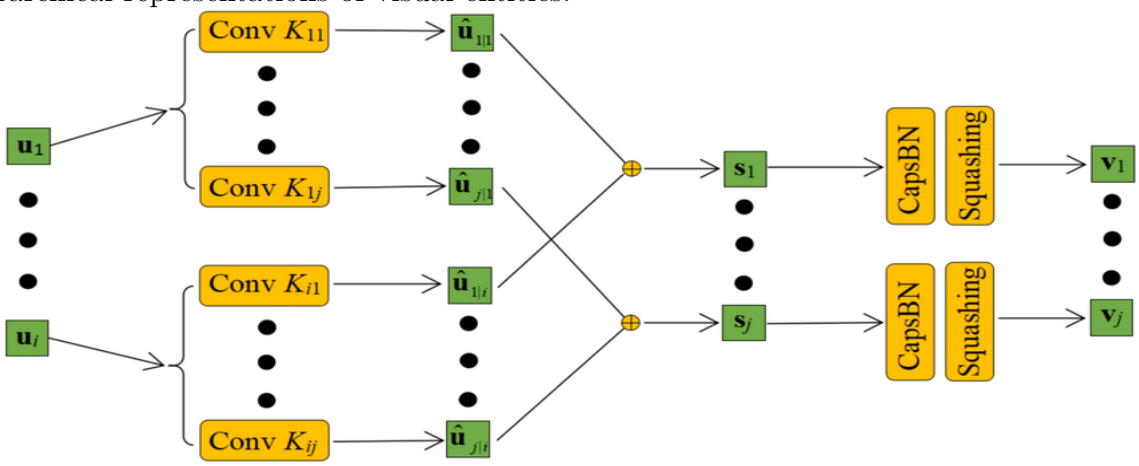
6. Agreement and convergence: The iterative routing process continues for a fixed number of iterations or until convergence is achieved. The capsules aim to agree on the instantiation parameters, existence, and properties of entities within the input data.

By iteratively refining the coupling coefficients and recalculating the prediction vectors, the routing-by-agreement mechanism enables capsules in different layers to collaborate and establish a consensus on the features and entities present in the input data. This consensus contributes to the hierarchical representation of complex relationships, providing a more robust and informative understanding of the input data within the CapsNet architecture.

CapsNet includes a reconstruction network that aims to reconstruct the input image from the higher-level capsule activations. The reconstruction network helps in training the CapsNet by providing a regularization mechanism. The model can learn to preserve the essential features during the routing process by comparing the reconstructed image with the original input. The CapsNet architecture offers several advantages over traditional CNNs. It can capture more nuanced features, handle spatial variance and deformation, and provide interpretability by encoding instantiation parameters. In the context of pneumonia diagnosis from chest X-rays, the CapsNet architecture can extract meaningful features and patterns related to pneumonia presence, enhancing the accuracy and efficiency of automated image classification. By leveraging the spatial relationships and hierarchical representations captured by the CapsNet, the model can provide more reliable and interpretable predictions for pneumonia diagnosis.

3.6.3 Mathematical framework

In the context of the CapsNet architecture, let's now dive into the mathematical equations that govern its statistical framework. These equations represent the transformations and computations within the primary capsule layer, digit capsule layer, and routing algorithm. By understanding these equations, we can learn how the CapsNet architecture learns hierarchical representations of visual entities.



The input into CapsNet is the actual image supplied to the neural net. The input image is three-dimensional, with pixel values representing the height and width of the image, and the 3rd dimension contains the color channels. The chest X-ray images were RGB with 224 by 224 pixels. This was captured in the input and stored in a 224x224x3 matrix [224, 224,

3]. Let X represent the input image tensor received by the input layer as raw data, in this case, the chest X-ray images. The input X is fed into the primary capsule layer.

The primary capsule layer aims to generate primary capsules representing low-level features extracted from the input image. It extracts features from the image vector by convolution. In image processing and computer vision, convolution applies a filter or kernel to an input image to extract specific features or information. Consider an image I which can be represented as n_1 by n_2 array and defined by a function as follows;

$$I : \{1, ..., n_1\} * \{1, ..., n_2\} \to W \subseteq \mathbb{R}(I, j) \mapsto I_{i,j} \tag{3.1}$$

Consider also a given Gaussian filter K;

$$K \in \mathbb{R}^{2h_1=1*2h_2+1} \tag{3.2}$$

$$= \begin{Bmatrix} K_{-h_1,-h_2} & ... & K_{-h_1,h_2} \\ ... & K_{0,0} & ... \\ K_{h_1,-h_2} & ... & K_{h_1,h_2} \end{Bmatrix} \tag{3.3}$$

The discrete convolution of the image I given filter K is:

$$(I * K)_{r,s} := \sum_{u=-h_1}^{h_1} \sum_{v=-h_2}^{h_2} K_{u,v} I_{r+u,s+v} \tag{3.4}$$

The convolution is followed by the rectified linear activation function (ReLU), a non-linear function that outputs the input directly if it is positive; otherwise, it outputs zero. It is defined as;

$$f(x) = \mathbf{max}(0, x) \tag{3.5}$$

The output from convolution is then captured in the primary capsule, which also works on the basic idea of convolution but captures complex features from the stack of convolution outputs. It has a defined number of primary capsules whose job is to take basic features detected by the convolution and produce combinations of these features. Capsules perform their computations on their inputs and then "encapsulate" the results into a small vector

of highly informative outputs. The output of the primary capsule can be represented as a combination of vectors u_i. These vectors encode probabilities that lower-level capsules detected in the corresponding objects, and directions of the vectors encode some internal state of the detected objects. This is now fed into the dense capsule layer.

The dense capsule layer takes the primary capsules as input and models higher-level features by learning the instantiation parameters of the capsules. The input u_i is the capsule unit. In the dense capsule layer, the following procedures take place;

1. Matrix multiplication of input vectors-The vectors u_i are multiplied by corresponding weight matrices W_{ij} that encode important spatial and other relationships between lower-level features and higher-level features. This is done such that the vectors achieve the affine transformation. This multiplication gives the prediction vector $\hat{\mu}_{j|i}$ as follows;

$$\hat{\mu}_{j|i} = W_{ij} u_i \tag{3.6}$$

2. Scalar weighting of input vectors-here lower level capsules need to "decide" to which higher level capsule to send their output. To make this decision, the weights are adjusted by a coupling coefficient C_{ij}, which is introduced to multiply a capsule's output by the prediction vector. The decision the capsule makes relies on the dynamic routing algorithm. The prediction vector is multiplied by the corresponding coupling coefficient C_{ij};

$$s = C_{ij} \hat{\mu}_{j|i} \tag{3.7}$$

3. Sum of weighted input vectors- The weighted input vectors are then combined to give the output s_j:

$$s_j = \sum C_{ij} \hat{\mu}_{j|i} \tag{3.8}$$

4. Non-linear activation function- The CapsNet model introduced "Squashing", a novel nonlinear activation function that takes a vector and then "squashes" it to have a length of no more than 1 but does not change its direction. The squashing function is given by;

$$a \Leftarrow \frac{\|a\|^2}{\|a\|^2} \frac{a}{\|a\|} \tag{3.9}$$

The Squashing function is used to transform vector s_j into vector v_j;

$$v_j = \frac{\|s_j\|^2}{\|s_j\|^2} \frac{s_j}{\|s_j\|} \tag{3.10}$$

Dynamic routing

The essence of the dynamic routing algorithm is that a lower-level capsule will send its input to the higher-level capsule that "agrees" with its input. The algorithm updates the dense capsule layer's coupling coefficient C_{ij}. The algorithm determines the modification of weights in a network, defining the strategy for assigning weights to the neurons' connections. A capsule network adjusts the weights such that a low-level capsule is strongly associated with high-level capsules in its proximity. The affine transformation determines the proximity measure. The distance between the outputs obtained from the affine transformation step and the dense clusters of the predictions of low-level capsules is computed. Dense clusters are formed if the predictions made by the low-level capsules are similar, thus lying near each other. The high-level capsule with the minimum distance between the cluster of already made predictions and the newly predicted one will have a higher weight, and the remaining capsules will be assigned lower weights based on the distance metric. The procedure is as follows;

Procedure 1 Routing algorithm.

procedure ROUTING($\hat{\mathbf{u}}_{j|i}, r, l$)
  for all capsule i in layer l and capsule j in layer $(l+1)$: $b_{ij} \leftarrow 0$.
  for r iterations **do**
    for all capsule i in layer l: $\mathbf{c}_i \leftarrow \texttt{softmax}(\mathbf{b}_i)$ ▷ softmax computes Eq. 3
    for all capsule j in layer $(l+1)$: $\mathbf{s}_j \leftarrow \sum_i c_{ij}\hat{\mathbf{u}}_{j|i}$
    for all capsule j in layer $(l+1)$: $\mathbf{v}_j \leftarrow \texttt{squash}(\mathbf{s}_j)$ ▷ squash computes Eq. 1
    for all capsule i in layer l and capsule j in layer $(l+1)$: $b_{ij} \leftarrow b_{ij} + \hat{\mathbf{u}}_{j|i}.\mathbf{v}_j$
  return $\mathbf{v}_j$

The input of the dynamic routing algorithm is the prediction vector $\hat{\mu}_{j|i}$ and the number r of routing iterations. The procedure can be described as follows:

1. This defines the procedure of ROUTING, which takes affine transformed input $\hat{\mu}_{j|i}$, the number of routing iterations r, and the layer number l as inputs.

2. Introduce b_{ij} a temporary value used to initialize the coupling coefficient C_{ij}.The value will be updated in the iteration process.

3. The for loop iterates r times.

4. The softmax function applied to b_{ij} makes sure to output a non-negative coupling coefficient C_{ij}, where all the outputs sum to 1 as follows:

$$C_{ij} = \frac{\mathbf{e}^{b_{ij}}}{\sum_k \mathbf{e}^{b_{ik}}} \tag{3.11}$$

5. The weighted sum is computed for every capsule in the succeeding layer.

6. The weighted sum is squashed for every capsule in the succeeding layer.

7. The weights b_{ij} are checked and updated.

$$b_{ij} \leftarrow b_{ij} + \hat{\mu}_{j|i} * v_j \tag{3.12}$$

$\hat{\mu}_{j|i}$ denotes the input to the capsule from low-level capsule$i$, and v_j denotes the output of high-level capsule j.

Once the weights are updated, the algorithm returns and restarts the c_{ij} calculation and repeats r times. After r times, all outputs for higher-level capsules were calculated, and routing weights were established. The forward pass can continue to the next level of the network. The dense layer will produce a set of capsule activations of equal dimensions. The activations represent the "explicit pose parameters" of the class each chest X-ray image is associated with. The output layer performs classification based on the presence and properties of the entities represented by the dense capsules. The probabilities of the different classes are obtained through a softmax activation function:

$$y = softmax(v_j) \tag{3.13}$$

$$y = \frac{\mathbf{e}^{v_j}}{\sum_k \mathbf{e}^{v_j}} \tag{3.14}$$

The loss function

The loss function used for training the CapsNet is a popular variation of "Hinge Loss", namely "Squared Hinge loss." The basic idea is that the model "wants" the correct class for each image to score higher than the incorrect classes by some fixed margin. It is represented as follows;

$$L_c = T_c(\mathbf{max}(0, m^+ - \|V_c\|)^2 + \lambda(1 - T_c)\mathbf{max}(0, \|V_c\| - m^-)^2 \tag{3.15}$$

Where:

$T_c = 1$ if a digit of class k is present

$m^+ = 0.9$

m^-- $= 0.1$

v_j = vector obtained from dense capsule layer

The first term of the equation represents the loss for a correct capsule, and the second term represents the loss for an incorrect capsule. The value of λ is 0.5 to reduce the loss when some categories do not appear, to ensure that the vector modulus representing the digital capsules of this category is as large as possible, and the vector modulus of other categories is as small as possible to achieve accurate classification. The total loss is the sum of all the digital capsule losses.

3.6.4 Merits of Capsnet model

The CapsNet model offers several advantages compared to traditional convolutional neural networks (CNNs) in various applications:

- CapsNet introduces the routing-by-agreement mechanism, enabling capsules in different layers to communicate and establish agreement on the instantiation parameters and existence of entities. This mechanism allows for the hierarchical representation

of features and spatial relationships, capturing more complex patterns and enabling a better understanding of the input data.

- CapsNet exhibits better resilience to variations in viewpoint and occlusion than CNNs. The capsule-based architecture allows capsules to encode pose information, facilitating the model's ability to handle different viewpoints and partial occlusions of objects.

- The use of capsules in CapsNet provides more interpretable representations of features and entities. Each capsule represents a specific feature or entity, encoding its presence, properties, and spatial relationships. This makes understanding and interpreting the model's inner workings easier than traditional CNNs.

- CapsNet has shown improved generalization capabilities, reducing the risk of overfitting. By capturing hierarchical relationships and utilizing dynamic routing, CapsNet allows for better generalization to new, unseen data.

- CapsNet has demonstrated promising performance on various challenging tasks, including object recognition, image segmentation, and pose estimation. Its ability to capture intricate spatial relationships and handle occlusion improves performance in complex real-world scenarios.

- CapsNet's hierarchical representation and routing mechanism can enable better learning from limited labeled data. The architecture's ability to capture informative features and relationships may reduce the reliance on extensive labeled datasets, making it advantageous in scenarios with limited labeled data availability.

3.7 Training the model

The most crucial phase of machine learning is training. To detect patterns and generate predictions, you feed your machine learning model the prepared data during training. As a result, the model gains knowledge from the data to complete the given task. The model improves in prediction over time with training. Training of the model involves the following steps and procedures:

1. Network initialization: Initialize the CapsNet architecture parameters, including the capsules' weights and any other learnable parameters in the model. The Xavier/Glorot initialization, designed to maintain the variance of activations and gradients throughout the network, was used in this study. It sets the initial weights of the parameters based on the size of the input and output dimensions of each layer. It helps prevent the issue of vanishing or exploding gradients, which can hinder the training process. The Xavier initialization is typically applied to the weights of the capsules in the network. The biases can be initialized to small values, such as zeros or a small constant, as they impact the network's overall performance less.

2. Forward propagation: Pass the training data through the network in a forward propagation step. Calculate the activations and outputs of the capsules in each layer based on the input data and the current values of the parameters. The squashing function is specifically designed to activate capsules in the CapsNet architecture. The squashing activation function takes the output vector of a capsule and squashes it to a length between 0 and 1 while maintaining the direction or orientation of the vector. The purpose of the squashing function is to represent the probability or likelihood of the presence of a particular entity or feature.

3. Loss calculation: Define a suitable loss function that quantifies the discrepancy between the model's predictions and the true labels. Standard loss functions used in classification tasks include cross-entropy loss or margin loss. The cross-entropy loss function was used as it is widely used for classification tasks. It measures the dissimilarity between the predicted probabilities and the true labels. Cross-entropy loss encourages the predicted probabilities to align with the true labels, thus guiding the model toward accurate classification.

4. Backpropagation and gradient update: Perform backpropagation to compute the gradients of the loss with respect to the model's parameters. Update the parameters using an optimization algorithm, such as stochastic gradient descent (SGD) or its variants, by adjusting the parameters in the direction of the negative gradients.

5. Iterative training: Repeat steps 3-5 for a predefined number of iterations or epochs.

The network learns from the training data in each iteration by adjusting the parameters based on the gradients and the optimization algorithm.

6. Validation and hyperparameter tuning: Periodically evaluate the model's performance on a separate validation dataset. Monitor accuracy, precision, recall, and other relevant measures. Use the validation results to fine-tune hyperparameters, such as learning rate, regularization parameters, or network architecture, to improve the model's performance.

7. Testing: Once complete training, assess the model's performance on a separate testing dataset not used during training or validation. This provides an unbiased evaluation of the model's generalization ability and performance on unseen data.

The study also employed the following techniques in training:

- Batch processing can make Neural networks faster, more effective, and more stable. It is essential to perform batch normalization, mainly when dealing with deep learning models. Batch normalizing is added to the closest to the output layer. It enables the achievement of faster convergence. Training is typically performed over multiple epochs, where one epoch represents one complete iteration through the entire training dataset. Data is processed in batches during each epoch to speed up training and utilize computational resources efficiently.

- Training a deep neural network takes much time and effort, especially when dealing with difficult situations. However, we may reduce the time required by selecting the appropriate optimizer. The Adam optimizer was used for this study.

- Hyper-parameters tuning is critical since hyper-parameters directly control the behavior of the training algorithms, having an essential impact on the model's performance under training. They determine how perfectly the model is going to perform. The main hyper-parameters are the learning rate, number of epochs, hidden layers, and activation functions.

- Regularization with reconstruction was also utilized in training. The reconstruction loss is computed by comparing the output of the reconstruction network with the

original input images. The combined loss of the classification and reconstruction tasks is used to update the model's parameters.

3.8 Model accuracy

This study used the following statistical metrics to evaluate the accuracy and performance of the CapsNet model in diagnosing pneumonia. Let;

TP represents the number of correctly predicted positive instances.

FP represents the number of instances predicted as positive but actually negative.

TN represents the number of correctly predicted negative instances.

FN represents the number of instances predicted as negative but actually positive.

- Accuracy measures the overall correctness of the model's predictions by calculating the ratio of correctly classified instances to the total number of instances. It indicates the proportion of correctly classified instances out of all instances in the dataset. Higher accuracy values indicate a better-performing model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.16}$$

- Precision measures the proportion of correctly classified images from the total number of images classified as a specific class. It indicates the model's ability to minimize false positives.

$$Precision = \frac{TP + TN}{TP + FP} \tag{3.17}$$

- Recall calculates the proportion of correctly classified images out of the total number of actual instances of ground truth positives. Recall gives insights into the model's ability to identify positive instances correctly.

$$Precision = \frac{TP}{TP + FN} \tag{3.18}$$

‘

- The F1 Score is the harmonic mean of precision and recall, providing a balanced

evaluation metric. It considers both false positives and false negatives, making it useful in scenarios where the dataset is imbalanced.

$$F1score = \frac{2 * (Precision * Recall)}{Precision + Recall} \tag{3.19}$$

- A confusion matrix is a table that summarizes the model's performance by showing the number of true positives, true negatives, false positives, and false negatives. It offers a comprehensive view of how the model performs across different classes and can aid in identifying specific areas where the model may struggle.
- Receiver Operating Characteristic (ROC) curve: The ROC curve is a graphical representation of the model's performance at various classification thresholds. It plots the true positive rate (recall) against the false positive rate, providing insights into the trade-off between true positives and false positives.
- Area under the curve (AUC): The AUC is a metric used to quantify the overall performance of a classification model based on the ROC curve. It represents the percentage of correctly classified images and provides a single value that summarizes the model's performance across all thresholds.

3.9 Model comparison

There was interest in comparing the performance of the proposed CapsNet model with different powerful deep-learning models; four variations were considered. Each model was trained and evaluated using the same dataset and experimental setup.

Model A was a convolutional neural network (CNN) architecture consisting of multiple convolutional layers followed by fully connected layers. Dropout regularization was applied to mitigate overfitting. The model was trained using the Adam optimizer with a learning rate of 0.001.

Model B utilized transfer learning with a pre-trained ResNet-50 model as the backbone. The pre-trained weights were fine-tuned on the pneumonia classification task. The fully connected layers on top of the ResNet-50 backbone were modified to match the number of classes. The model was trained using the Adam optimizer with a learning rate of 0.001.

Model C combined the strengths of CNN and LSTM architectures to form an ensemble model. It incorporated convolutional layers to extract spatial features from the images and LSTM layers to capture sequential dependencies. The individual models were trained independently, and their predictions were combined using a voting mechanism. The ensemble model aimed to leverage the complementary strengths of CNN and CNN-LSTM architectures for improved performance.

Model D was an autoencoder model

Each model was trained for 50 epochs with a batch size of 32. The performance of each model was evaluated using accuracy, precision, recall, and F1 score as the evaluation metrics.

Chapter 4

RESULTS AND FINDINGS

4.1 Introduction

This section entails the study's results and findings on the early diagnosis of pneumonia from chest X-rays using a CapsNet model. It provides a detailed analysis and interpretation of the data, highlighting the performance of the CapsNet model in accurately identifying pneumonia cases. The primary objective of this study was to develop and evaluate a CapsNet model for the early detection of pneumonia from chest X-ray images. The model was trained on a carefully curated dataset comprising pneumonia-positive and pneumonia-negative cases, and its performance was assessed using various evaluation metrics. Furthermore, a comparison was made between the CapsNet and baseline models and approaches commonly employed in pneumonia diagnosis from chest X-rays. This analysis aimed to demonstrate the superiority of the CapsNet model and its potential to outperform existing methods.

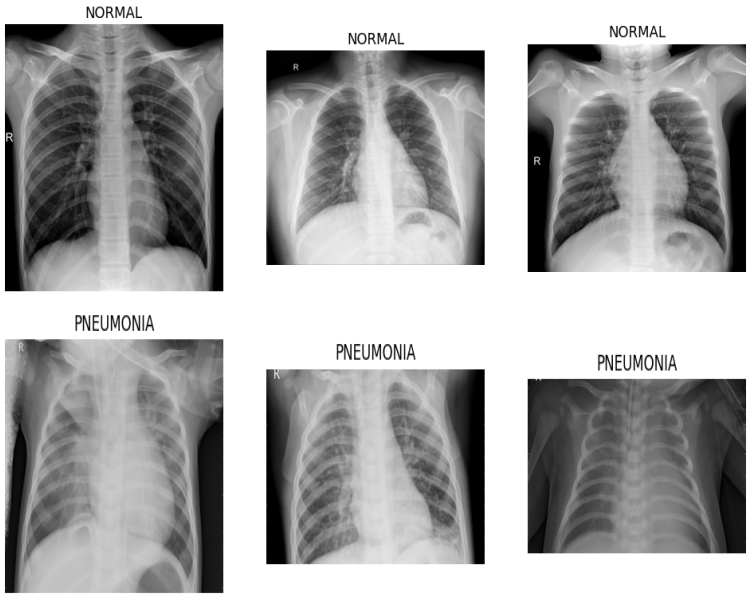
4.2 Dataset

The dataset used in this study was secondary data obtained from Mendeley data, developed and uploaded by Kermany et al. (2018a) in 2018. The dataset consisted of 5,856 chest X-ray images. Among these, 1,583 were normal chest X-ray images from patients without pneumonia, while 4,273 were pneumonia-infected chest X-ray images from patients infected with pneumonia. The distribution in the dataset reflects the prevalence of pneu-

monia cases in the dataset. It ensures that both normal and pneumonia-positive cases are adequately represented for training and evaluation. The dataset's composition enabled the CapsNet model to learn and differentiate between normal and pneumonia cases effectively. Appropriate pre-processing techniques were implemented.

4.3 Data pre-processing

Data pre-processing was a major step in analyzing the chest x-ray images. It was essential to ensure the achievement of the study's main goal. The chest X-ray images were viewed as follows for both normal and pneumonia cases:

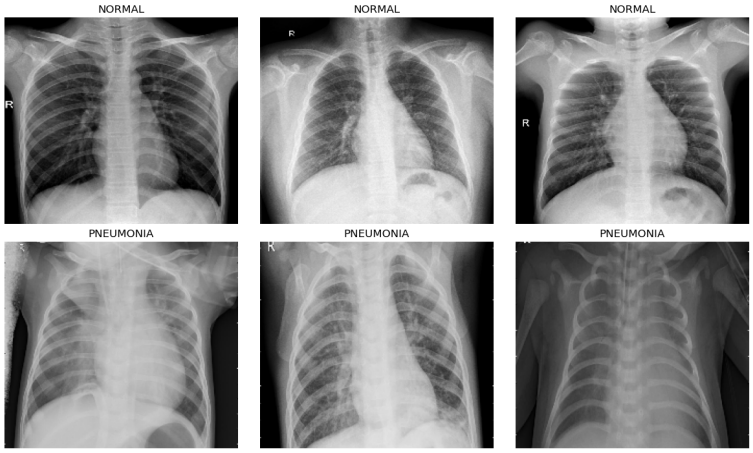


The images were in grayscale format with randomly varying image sizes and shapes. Grayscale images are focused more on the image's structural aspects, reducing the complexity of the data and making it easier for the model to learn and extract meaningful features. The image scale was in the range (0,255) for all the images in the dataset. The images were subjected to the following pre-processing procedures.

4.3.1 Image resizing

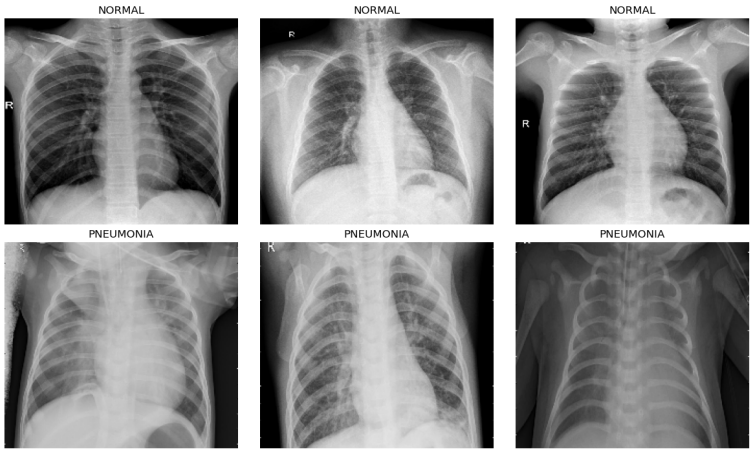
The images were resized to change their dimensions, i.e., width and height. This step ensured that all images in the dataset had the same dimensions, making them compatible

for further processing and analysis. The images that were resized and reshaped can be viewed as follows:



4.3.2 Image standardization

The images were then standardized to ensure that the pixel values within each image were transformed to a consistent scale, making them suitable for analysis. This step helped to standardize the input size for the model. The images' pixel values were normalized to a standard range, typically between 0 and 1, by min-max normalization. This helped to bring all the pixel values to a similar scale and improved the model's convergence during training. The standardized images can be viewed as follows:



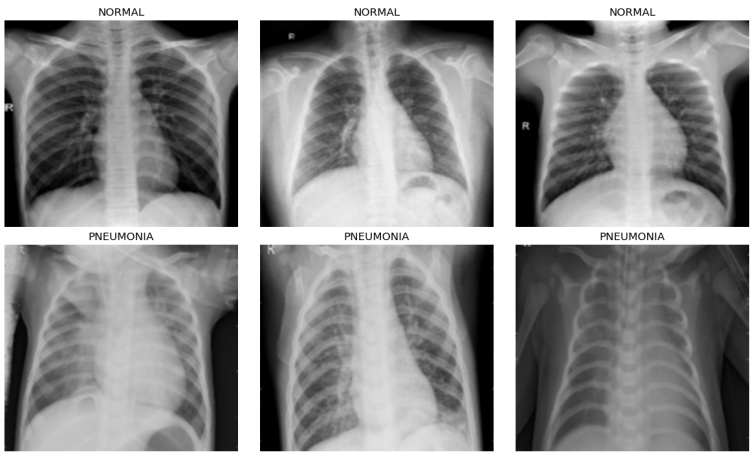
4.3.3 Noise reduction

Noise reduction aims to remove and reduce unwanted noise from images. Noise can arise from various sources, such as sensor limitations, transmission errors, or environmen-

tal factors. Removing noise helps improve image quality, enhance details, and facilitate more accurate analysis and feature extraction. Gaussian blur was applied, a widely used image-processing technique for reducing noise and smoothing images. It is based on the mathematical concept of convolving an image with a Gaussian kernel.

Gaussian blur operates by applying a Gaussian kernel to each pixel in the image. This kernel is a weighted matrix that assigns higher weights to central pixels and lower weights to surrounding pixels. The weights are determined by the Gaussian distribution, creating a bell-shaped curve. Gaussian blur effectively reduces random and high-frequency noise in an image. Random noise, such as electronic noise in photographs, is effectively smoothed out. High-frequency noise, which manifests as small, rapid variations in pixel values, is also mitigated. When the Gaussian kernel is convolved with the image, each pixel's value is replaced by a weighted average of its neighbors' values. This averaging effect helps reduce the impact of noisy pixel values, resulting in a smoother appearance.

The images were viewed as follows:

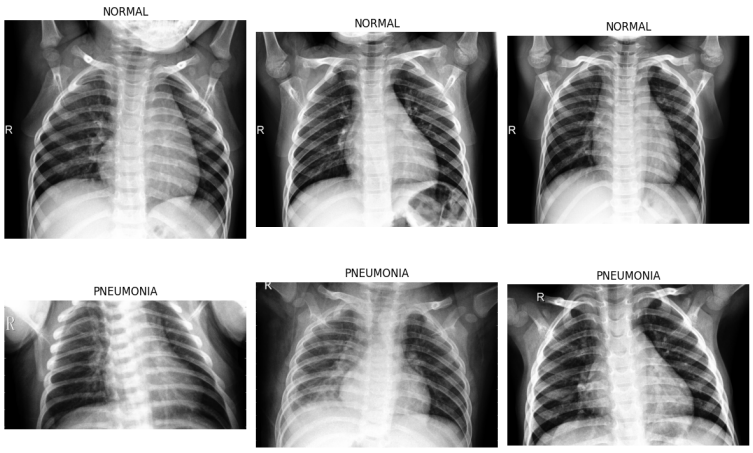


4.3.4 Contrast enhancement

Contrast enhancement is a technique used to improve images' visual quality and distinguishability by increasing the contrast between different pixel intensities. It aims to enhance the differences in brightness or color between image regions, making them more prominent and improving the perception of details. Histogram equalization was applied to the images; it is a widely used image enhancement technique that helps improve an image's contrast and visibility. It redistributes the pixel intensities in an image to cover a broader range of values, enhancing its overall appearance. Histogram equalization was applied to

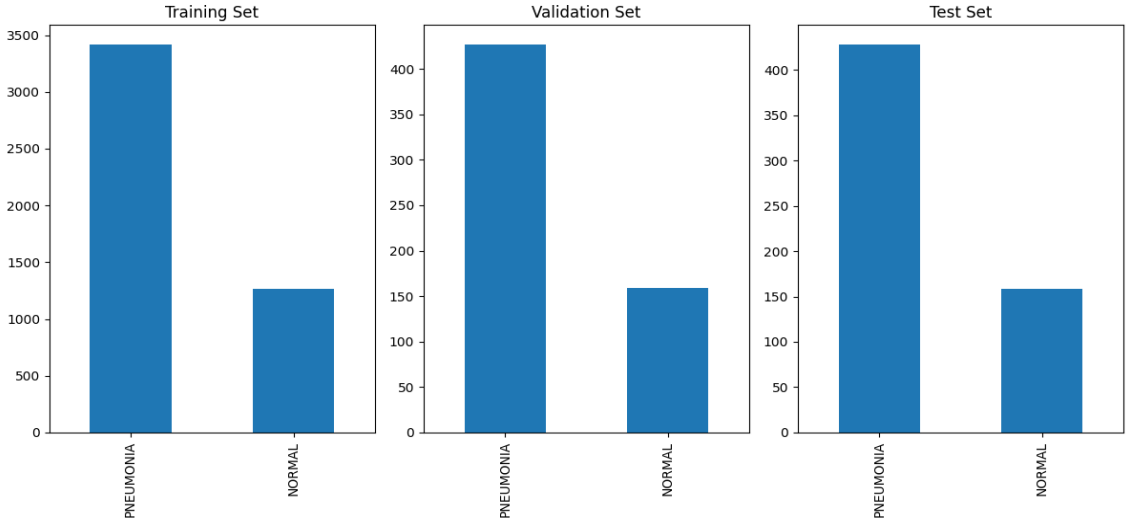
the images to help improve the visibility of details to capture the presence of any pneumonia infiltrations. The primary goal of histogram equalization is to make an image's histogram as uniform as possible. This means that all intensity levels within the image will be equally represented, resulting in enhanced contrast.

Histogram equalization can significantly enhance the contrast in an image. It stretches the range of intensity levels, making dark and bright areas darker, resulting in a more visually appealing image. It is used in medical imaging to enhance details in X-ray and MRI images. The enhanced images were viewed as follows:



4.3.5 Data partitioning

The images were then split into 80% to 20% proportions for model training and evaluation, respectively. The 80% proportion was used as the training set, while 10% was used as the validation set and 10% as the testing set. There were 4684 chest x-ray images in the training set and 586 in the validation and testing sets. They were distributed as follows:



The distribution was imbalanced for the normal and pneumonia cases. Still, the split ensured a balanced distribution of proportions of cases in all the sets, which is key in the model validation and evaluation after training.

4.4 Model training

The next step was to proceed with model training. In the pre-processing phase, we took several important steps to ensure the robustness of our model. To wrap up the pre-processing, shuffling the proportional files was included, a critical process to guarantee that our model would be capable of making accurate predictions across a wide range of data. This shuffling was accomplished using Python's TensorFlow libraries. Additionally, we carefully randomized the testing dataset to assess our model's ability to handle new and previously unseen data, minimizing any interference from previously encountered data points.

The Capsule Neural Network (CapsNet) was meticulously trained throughout 50 epochs, each processing a batch size 32. A learning rate of 0.001 was employed throughout the training process. The output below was obtained to mark the training process:

```
_____________________________________________
Epoch 1/50
147/147 [==============================] - 422s 3s/step - loss: 0.3736 - accuracy: 0.8405 - val_loss: 0.2962 - val_accuracy: 0.8703
Epoch 2/50
147/147 [==============================] - 433s 3s/step - loss: 0.2795 - accuracy: 0.9167 - val_loss: 0.2671 - val_accuracy: 0.9454
Epoch 3/50
147/147 [==============================] - 423s 3s/step - loss: 0.2426 - accuracy: 0.9387 - val_loss: 0.2358 - val_accuracy: 0.9420
Epoch 4/50
147/147 [==============================] - 429s 3s/step - loss: 0.2252 - accuracy: 0.9430 - val_loss: 0.2204 - val_accuracy: 0.9590
Epoch 5/50
147/147 [==============================] - 415s 3s/step - loss: 0.2027 - accuracy: 0.9535 - val_loss: 0.1976 - val_accuracy: 0.9573
Epoch 6/50
147/147 [==============================] - 427s 3s/step - loss: 0.1792 - accuracy: 0.9599 - val_loss: 0.2019 - val_accuracy: 0.9437
Epoch 7/50
147/147 [==============================] - 414s 3s/step - loss: 0.1423 - accuracy: 0.9530 - val_loss: 0.1223 - val_accuracy: 0.9505
Epoch 8/50
147/147 [==============================] - 413s 3s/step - loss: 0.0822 - accuracy: 0.9716 - val_loss: 0.1325 - val_accuracy: 0.9505
Epoch 9/50
147/147 [==============================] - 415s 3s/step - loss: 0.0596 - accuracy: 0.9787 - val_loss: 0.1235 - val_accuracy: 0.9556
Epoch 10/50
147/147 [==============================] - 407s 3s/step - loss: 0.0406 - accuracy: 0.9855 - val_loss: 0.1758 - val_accuracy: 0.9471
```

During training, several noteworthy observations were made. Initially, model accuracy was significantly improved as the number of epochs increased, peaking at epoch 18. Beyond this point, the model's accuracy reached a plateau, indicating that further epochs did not significantly enhance its performance. Interestingly, it was also observed that the time required for each successive training phase decreased as we progressed through the epochs.

Additionally, the model demonstrated its proficiency in minimizing the loss function, which consistently decreased with each epoch. This pattern indicated that the model effectively penalized deviations from the expected outcomes, reflecting a steady linear learning rate.

These findings collectively underscore the successful training of our CapsNet model, which exhibited substantial improvements in accuracy and loss reduction over the training epochs. These outcomes lay the groundwork for evaluating the model's efficacy in diagnosing pneumonia from chest X-ray images.

4.5 Model evaluation

The model performance was evaluated using the following metrics:

4.5.1 Accuracy

Accuracy indicates the proportion of correctly classified images out of all images in the dataset. This study achieved accuracy as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = 0.96 \tag{4.1}$$

An accuracy of 0.96 indicates that the model correctly predicted the class of 96% of the samples in the dataset, which is a high overall correctness rate. This implies the CapsNet model performed well in diagnosing pneumonia from the chest X-ray images.

4.5.2 Precision

Precision indicates the CapsNet model's ability to minimize false positives. The precision was:

$$Precision = \frac{TP + TN}{TP + FP} = 0.9697 \tag{4.2}$$

A precision of 0.9697 suggests that 96.97% of the predicted pneumonia cases were indeed true positives, implying a reasonably low rate of false positive predictions.

4.5.3 Recall

Recall gives insights into the model's ability to identify positive instances correctly.

$$Precision = \frac{TP}{TP + FN} = 0.9742 \tag{4.3}$$

‘ A recall of 0.9742 means that the model correctly captured 97.42% of the positive cases, indicating that it can detect pneumonia cases.

4.5.4 F1 score

The model F1 score considers both false positives and false negatives, as follows:

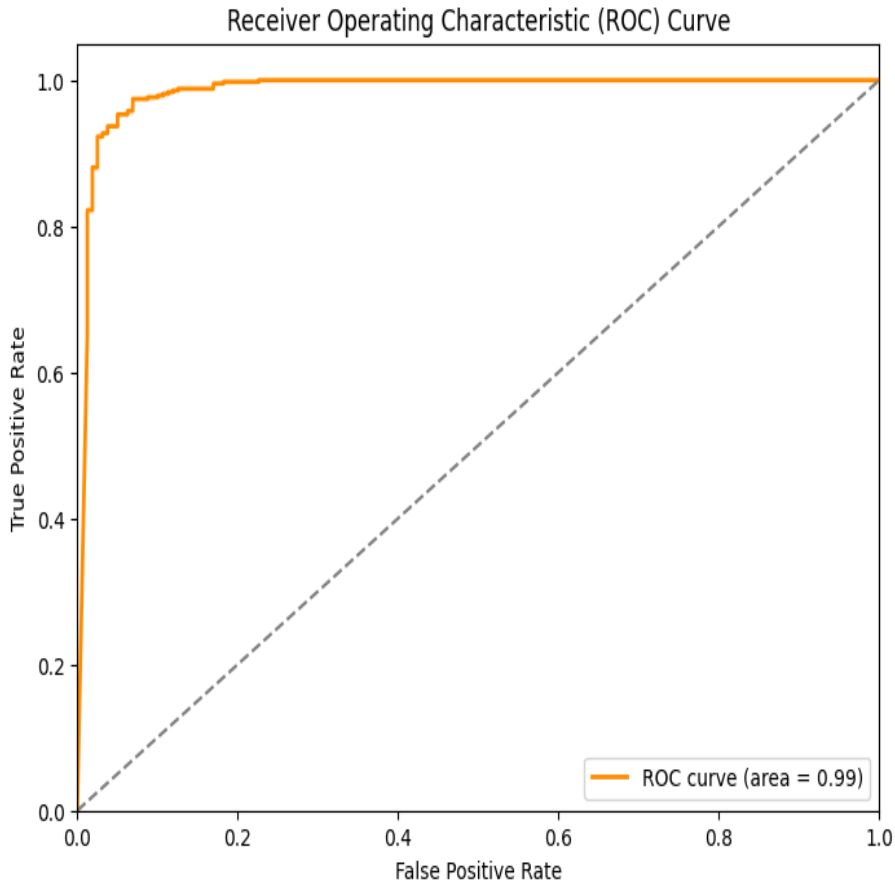
$$F1score = \frac{2 * (Precision * Recall)}{Precision + Recall} \tag{4.4}$$

$$= \frac{2 * (0.9697 * 0.9742)}{0.9697 + 0.9742} = 0.972 \tag{4.5}$$

With an F1 score of 0.972, the model demonstrates a good trade-off between precision and recall, indicating overall effective performance.

4.5.5 ROC curve

The ROC curve is a graphical representation of the model's performance at various classification thresholds. The ROC curve or the CapsNet was as follows:



The ROC curve above indicates higher true positives and lower false negatives because there are large values in the y-axis, which indicates the true positive rate. The curve for classification was deeply elongated to the top left, indicating a high performance for the classification model. The area under the curve for the classification was also obtained from the curve. The AUC was 0.99, which implies outstanding discrimination in the classification. The CapsNet model performed well in classifying pneumonia cases, making it very suitable for early pneumonia diagnosis.

4.6 Model comparison

The study also explored and compared the findings with other deep-learning models such as Autoencoder, CNN model, a hybrid CNN-LSTM model, and a ResNet transfer learning model. The following results were obtained:

Models/Metrics	Train accuracy	Test accuracy	Precision	Recall	F1 score	Run time (Mins)
CapsNet	0.9224	0.9678	0.8214	0.8423	0.8317	36.6
Autoencoder	0.9012	0.9212	0.7819	0.7422	0.7615	40.2
CNN	0.9002	0.9162	0.8022	0.7618	0.7814	48
CNN-LSTM	0.8812	0.8400	0.7814	0.7212	0.7501	62.3
ResNet	0.8406	0.8911	0.7224	0.7042	0.7132	56.2

The autoencoder achieved an accuracy of 0.9224, indicating that it correctly classified 92.24% of the samples in the dataset. This suggests that autoencoders have high overall correctness in predictions. The CNN model, on the other hand, achieved an accuracy of 0.9212. Although slightly lower than Model autoencoders and CapsNet, it still demonstrates a relatively high level of accuracy. The hybrid CNN-LSTM and ResNet models achieved an accuracy of 0.84 and 0.8911, respectively. Though these models performed beyond the standard threshold, their performance was slightly lower, with the CNN-LSTM model being the poorest.

The models achieved precision values ranging from 0.72 to 0.82, indicating that above 72% of the predicted pneumonia cases were true positives. The CapsNet achieved a precision of 0.8214, indicating a slightly higher rate of correctly predicted pneumonia cases. This suggests that CapsNet may be better able to classify positive cases accurately.

Examining the recall, the CNN achieved a recall of 0.7618, meaning it correctly captured 76.18% of the positive cases. On the other hand, the autoencoder achieved a recall of 0.7422, and the CNN-LSTM and ResNet achieved 0.7212 and 0.7042, indicating a slightly lower rate of correctly identifying positive cases. This suggests that the CapsNet model still stood out, having a better ability to detect pneumonia cases.

Similarly, the CapsNet model had the highest F1 score, but all the models' scores ex-

hibited a reasonable trade-off between precision and recall. Overall, comparing the models reveals that each model has strengths and weaknesses, but the CapsNet model achieved a higher performance than the competing models.

4.7 Conclusion

The results and findings highlight the effectiveness of the CapsNet model as the main model for early diagnosis of pneumonia from chest X-ray images. With its powerful architecture and advanced features, the CapsNet model performs better in accurately classifying pneumonia cases. The high accuracy, precision, recall, and F1 score achieved by the CapsNet model emphasize its potential as a robust tool for pneumonia diagnosis. These findings underscore the importance of leveraging innovative deep learning techniques, such as CapsNet, in developing cutting-edge automated image classification systems for improved early detection and treatment of pneumonia.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

The main objective of this study was to develop and train the CapsNet model as a tool for early diagnosis of pneumonia from Chest X-rays. The major objective was to ensure high performance and efficiency in image classification. The CapsNet model was trained to capture low-level quality features, overlapping features, and complex image occlusions.

5.2 Conclusions

This research has contributed to the understanding and advancement of early diagnosis of pneumonia using chest X-ray images. Based on the findings presented, the following conclusions can be drawn:

The CapsNet model has demonstrated its power and effectiveness in accurately classifying pneumonia cases from chest X-ray images. Its innovative architecture and features have shown promising results, achieving high accuracy, precision, recall, and F1 score values. This indicates its potential to assist healthcare professionals with efficient and reliable pneumonia diagnosis. The pre-processing techniques employed in the research, such as image resizing, normalization, augmentation, histogram equalization, and noise reduction, have significantly contributed to improving the performance of the models. These techniques

enhance the quality of the input data, reduce noise, and enhance important image features, leading to more accurate and robust classification results.

The evaluation metrics used, including accuracy, precision, recall, and F1 score, provide a comprehensive assessment of the models' performance. The combination of these metrics enables a balanced evaluation of the models' ability to classify pneumonia cases correctly while minimizing false positives and false negatives. These metrics also enabled proper comparison between models.

5.3 Recommendations

Based on the research findings, the following recommendations are proposed for further investigation and future work:

There is a need to enhance the generalizability of the models, it is recommended to explore larger and more diverse datasets. Including data from different sources and populations can help evaluate the models' performance in varied scenarios and improve their robustness.

Investigate the potential of utilizing pre-trained models as feature extractors for pneumonia classification. Transfer learning can leverage the knowledge learned from large-scale datasets to enhance the performance of the models on relatively smaller datasets. The novelty comes in extracting features from a dataset to ensure high-level accuracy before being fed to a network.

Conduct further studies to validate the performance of the models in real-world clinical settings. Collaborating with medical professionals and evaluating the models' effectiveness on patient data can help assess their practical utility and integration into the healthcare workflow. The major issue with health science is the clinical validation of developing technologies. There is, therefore, a dire need to come up with a way to ensure validation and further initiation.

The deep learning field has a swamp of upcoming future research that can enhance the capabilities and applicability of automated systems for early pneumonia diagnosis, ultimately contributing to improved patient care and outcomes. The findings provide valuable insights into the development and evaluation of models for early diagnosis of pneumonia

from chest X-ray images.

Expand dataset: Include more diverse and representative samples to improve generalizability. Transfer learning: Explore leveraging pre-trained models for enhanced performance and faster training. Explainability: Investigate methods to understand and interpret the models' decision-making process. Clinical validation: Validate models in real-world medical settings using patient data. Comparative studies: Compare models with other state-of-the-art approaches for benchmarking. Continuous updating: Monitor and update models with new data to ensure reliability over time. Standardization: Collaborate to establish protocols and benchmarks for evaluating pneumonia diagnosis models. Integration: Explore integrating models into existing healthcare systems or telemedicine platforms. Remote diagnosis: Support remote screening and detection, especially in underserved areas or during public health crises. These recommendations aim to advance pneumonia diagnosis, improve model performance, and enhance patient care and accessibility.

5.4 Areas for further studies

Advanced Deep Learning Architectures: Explore the application of more advanced deep learning architectures, such as attention-based models, graph neural networks, or transformers, to enhance the performance of pneumonia diagnosis models.

Multi-Modal Approaches: Investigate the integration of multiple imaging modalities, such as X-ray images and CT scans, or the combination of imaging data with clinical and patient information, to improve the accuracy and robustness of pneumonia diagnosis models.

Uncertainty Estimation: Develop techniques for estimating model uncertainty, such as Bayesian deep learning or Monte Carlo dropout, to provide confidence measures for model predictions. This can help understand the reliability of the model's diagnosis and aid in clinical decision-making.

Real-Time Diagnosis: Focus on developing real-time or near real-time models that can provide instant feedback to healthcare professionals, enabling prompt diagnosis and treatment planning.

Transferability to Other Respiratory Diseases: Explore the transferability of the developed models to other respiratory diseases with similar imaging characteristics, such as

tuberculosis or lung cancer. Investigate the potential for a unified diagnostic framework for multiple respiratory conditions.

Interpretability and Explainability: Further investigate methods to explain and interpret the decisions made by the models, especially in the medical domain, to increase transparency and trust in automated diagnosis systems.

Large-Scale Collaborative Studies: Conduct large-scale multi-center studies involving collaboration among healthcare institutions and researchers to validate the performance and generalizability of the models across different populations, demographics, and geographic regions.

Clinical Decision Support Systems: Develop clinical decision support systems that integrate the pneumonia diagnosis models with electronic health records and clinical guidelines, providing comprehensive support to healthcare professionals in diagnosis and treatment planning.

Privacy and Ethical Considerations: Address privacy concerns and ethical considerations related to using patient data in developing and deploying automated diagnosis systems, ensuring compliance with data protection regulations, and maintaining patient confidentiality.

Cost-Effectiveness Analysis: Perform cost-effectiveness analyses to evaluate the economic impact of implementing automated diagnosis systems for pneumonia, considering factors such as improved patient outcomes, reduced diagnostic delays, and resource optimization.

By pursuing these avenues of research, we can further advance the field of early pneumonia diagnosis and contribute to improved healthcare outcomes and patient care.

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